Finpact

Split bills, track expenses, and settle up effortlessly!

Software Requirements Specification (SRS)

Phase (2)

CSAI_203_Course_Project

Introduction to Software Engineering

Zewail City of Science and Technology

School of Computational Sciences and Artificial Intelligence (CSAI)

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Disclaimer

The members of the FinPact team have prior exposure to the software industry, so this is not the first Software Requirements Specification we have written. The structure and level of detail follow the IEEE Std 830 practice we have applied before, adapted to the CSAI 203 Phase 2 guidelines and template.

Each detail of this document we have paid a lot of time and effort to make it perfect as much as we can within the available time frame and course context.

FinPact is an academic project. All names, amounts, and transactions used in examples are fictional. The system does not move real money and is not connected to any bank or payment provider; payments and settlements are informational records only. Stakeholder rows marked “Launch” in Section 2.4 describe a possible commercial release and are outside the Phase 2 scope.

All figures in this document (architecture, use-case, class, and state diagrams) were drawn in draw.io; the editable source file (FinPact_SRS_Diagrams.drawio) is kept in the repository’s /docs folder.

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1. Introduction:

1.1 Purpose:

In this Software Requirements Specification (SRS) we define the requirements for **"FinPact",** a web-based application that lets individuals track personal daily spending, split shared bills among roommates or team members, set monthly category limits, and view spending analytics. The main goal of this document is to provide a single, authoritative reference for designing, implementing, and testing the system, and also as a part of the CSAI 203 course project to verify that the delivered application satisfies the stated requirements.

Specifically, this SRS is used to:

- Specify the 20 functional requirements covering user accounts, personal expenses, shared expenses, settlements, budgets, and analytics.

- Define 5 testable non-functional requirements for performance, security, usability, reliability, and browser compatibility.

- Document the three use-case diagrams (Account/Personal Expenses, Shared Expenses/Settlements, and Budgets/Analytics) and their IEEE-template descriptions.

- Provide the conceptual class model (User, PersonalExpense, SharedExpense, Budget, Settlement, etc.) that guides the MVC design.

So, the document describes what FinPact shall do and the constraints under which it operates; it does not prescribe detailed implementation except where required by the course constraints (e.g., MVC).

1.2 Scope:

FinPact is a standalone, multi-user web application built for two overlapping groups: individuals who want to monitor their daily spending and monthly budgets, and groups (roommates, travel groups, project teams) who need to split bills and track who owes whom. The system records expenses, calculates each participant's share of a shared bill, tracks payments and outstanding balances, and records settlements without transferring real money.

1.2.1 In scope (Phase 2):

- User account management: registration, login, logout, and authenticated access control (FR-01–FR-02).

- Personal expense management: add, edit, delete, categorize, search/filter, and view personal expense history (FR-03–FR-08).

- Shared expense management: create shared bills, add participants, split bills (equal or custom), track payments/debts, and view shared history (FR-09–FR-13).

- Settlement management: record settlements between payer and receiver and update outstanding balances without performing real financial transactions (FR-20).

- Budget management: set monthly budgets, set category spending limits, monitor usage (spending, remaining, percentage), and trigger threshold warnings (FR-14–FR-17).

- Spending analytics: view category/period summaries and compare spending trends across periods (FR-18–FR-19).

- Web UI: responsive access through modern desktop and mobile browsers, with a digital user guide and FAQ.

1.2.2 Out of scope (Phase 2):

- Real money transfers, payment-gateway integration, or bank-account connections.

- Native iOS/Android applications.

- Public third-party APIs or integrations with accounting/tax software.

- Multi-currency conversion or investment tracking.

**The system is standalone and is not part of a larger product suite; it follows the MVC pattern and uses a relational database.

1.2.3 Target Usage Scenarios:

FinPact is designed around everyday scenarios, the following are just initial scenarios:

- A solo user records daily coffee and grocery expenses, sets a monthly Food category limit of EGP 2,000, and receives a warning when spending reaches 80% of that limit.

- Three roommates create a shared “October Rent” expense of EGP 9,000, split it equally, record who paid, and later record a settlement when one roommate transfers their share.

- A project team logs a shared dinner after a meeting, adds four participants, splits the bill, and tracks outstanding balances until everyone settles.

**These scenarios motivate the functional requirements in Section 3.1 and appear as the primary flows in the use cases in Section 3.2.2.

1.3 Definitions, Acronyms, and Abbreviations

Term/Acronym	Definition (FinPact-Specific)

Budget	A user-defined total spending limit for a specific monthly period (e.g., “October 2026: EGP 4,000”).
Budget Notification	A stored warning generated when spending reaches a threshold (e.g., 80% or 100%) of a budget or category limit.
Category Budget Limit	A spending limit assigned to one expense category (e.g., Food, Transport) within a monthly budget.
CRUD	Create, Read, Update, Delete, the operations applied to personal expenses, shared expenses, budgets, and settlements.
FinPact	The personal and shared expense and budget tracking application specified in this SRS.
FR	Functional Requirement: one of the 20 numbered requirements (FR-01–FR-20) in Section 3.1.
Guest	An unauthenticated visitor who can register or log in.
IEEE	Institute of Electrical and Electronics Engineers: source of the SRS outline and use-case description template.
MVC	Model-View-Controller: the required architectural pattern for FinPact
NFR	Non-Functional Requirement: one of the 5 quality requirements (NFR-01–NFR-05) in Section 3.4.
Participant	A registered user included in a shared expense; each participant has a recorded share amount.
Payment Record	A record of an amount paid by a user toward a shared expense.
Registered User	An authenticated FinPact account holder who can manage expenses, budgets, shared bills, and settlements.
Settlement	A recorded update indicating that a payer has settled an outstanding amount with a receiver for a shared expense; For now **FinPact does not transfer real money** (maybe later).
Shared Expense	A bill (e.g., rent, dinner, utilities) involving at least two participants, with shares, payments, and settlements tracked by FinPact.
UAT	User Acceptance Testing: testing used to confirm that FinPact meets the course project’s acceptance criteria.

1.4 References

[1] IEEE, “IEEE Recommended Practice for Software Requirements Specifications,” IEEE Std 830-1998, 1998.

[2] CSAI 203 - Introduction to Software Engineering, “Project Description and Phase 2 SRS Guidelines,” Zewail City of Science and Technology, School of Computational Sciences and Artificial Intelligence.

[3] IEEE-SRS Markdown Template, GitHub repository. [Online]. Available:

https://github.com/cemantix-org/ieee-srs/blob/master/IEEE-SRS.md

[4] PlantUML, “ Diagrams Documentation.” [Online]. Available: https://plantuml.com/use-case-diagram

(used to produce the three FinPact use-case diagrams in Section 3.2.1).

[5] CSAI 203 Course Lecture Slides, “UML Use Case IEEE Description Template,” Zewail City of Science and Technology (used for the UC-01–UC-20 descriptions in Section 3.2.2).

1.5 Overview

This document is organized into four sections. Section 1 introduces the purpose, scope, definitions, and references of the SRS. Section 2 provides an overall description of FinPact, including its product perspective, main functions, user classes, stakeholders, operating environment, design constraints, user documentation, and assumptions and dependencies. Section 3: specifies the detailed requirements: 20 functional requirements (FR-01–FR-20), use-case diagrams with IEEE-template descriptions (UC-01–UC-20), the conceptual class diagram and class descriptions, five non-functional requirements (NFR-01–NFR-05) with test methods and acceptance criteria, and external interface requirements. Section 4 contains Appendix A (Data Dictionary) and Appendix B (Glossary).

1.6 Document Conventions

Throughout this SRS:

- “Shall”: indicates a mandatory requirement that the delivered system must satisfy.

- “Should”: indicates a recommended requirement that may be waived only with documented justification.

- “May”: indicates an optional capability.

- Requirement identifiers (“FR-xx”, “NFR-xx”, “UC-xx”) are stable and are used for traceability into design, implementation, and testing artifacts.

- “FinPact” refers to the system specified in this document; “the system” is used interchangeably

2. Overall Description

This section describes FinPact's operating context, major functions, user classes, stakeholders, environment, constraints, user documentation, assumptions and dependencies, and apportioning of requirements. Consistent with IEEE Std 830-1998, this section supplies the background necessary to understand the requirements in Section 3 and does not itself introduce additional functional requirements. Where a statement in this section implies a constraint or assumption, the corresponding requirement is stated formally in Section 3 (FR-xx / NFR-xx) and is cross-referenced.

2.1 Product Perspective

FinPact is a standalone, self-contained system. It is not a replacement for an existing product, an upgrade, or a component of a larger product family. No legacy data migration or backward-compatibility requirements apply in Phase 2.

FinPact interacts with a small, well-defined set of external entities. Figure 1 shows the system context: users interact with FinPact exclusively through a web browser over HTTPS; the FinPact web server hosts the MVC application; and a relational database stores user, expense, shared-expense, budget, settlement, and notification records. No external payment gateway, banking API, email service, or third-party analytics platform is part of the Phase 2 boundary.

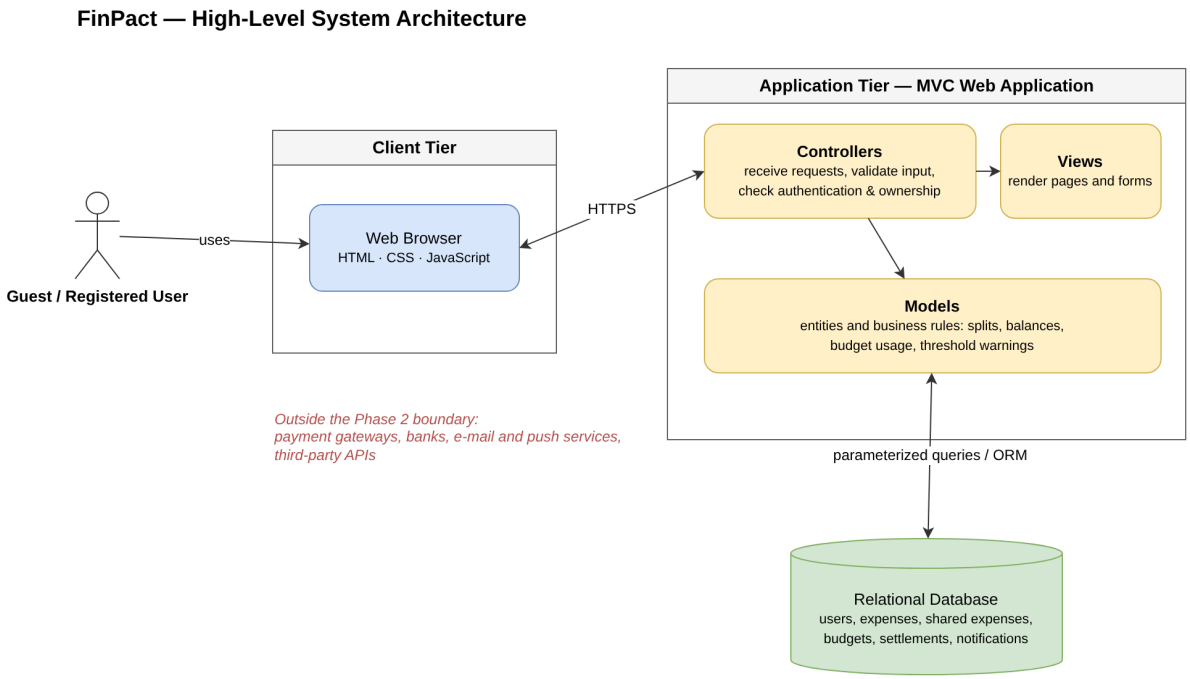


Figure 1: FinPact High-Level System Architecture

2.2 Product Functions:

At a high level, **FinPact provides six (6) functional areas (see the next table)**. Each area is refined into numbered requirements in Section 3.1. Priorities follow the convention High / Medium / Low and are used for planning, implementation ordering, and test coverage prioritization.

	Functional Area	Summary of Major Functions	Corresponding FRs	Priority
1	**User Account Management**	Registration, secure login, logout, and authentication of all protected pages.	FR-01, FR-02	High
2	**Personal Expense Management**	Add, edit, delete, categorize, search, filter, and view personal expenses.	FR-03 – FR-08	High
3	**Shared Expense Management**	Create shared bills, add participants, split bills (equal or custom), track payments and debts, view shared history.	FR-09 – FR-13	High
4	**Settlement Management**	Record settlements between payer and receiver and update outstanding balances without transferring real money.	FR-20	High
5	**Budget Management**	Set monthly total budgets, set category spending limits, monitor spending/remaining/usage percentage, receive threshold warnings.	FR-14 – FR-17	High
6	**Spending Analytics**	View category/period summaries and compare spending trends across periods.	FR-18, FR-19	Medium

**Deferred to future versions (not Phase 2): email or push notifications, recurring expenses, percentage- or weighted-based splits, multi-currency support, receipt image upload, CSV/PDF export, native mobile applications, external payment-gateway integration, and third-party API access.

2.3 User Classes and Characteristics

FinPact has three user classes. A registered user may belong to both registered classes simultaneously; the system does not require separate accounts for individual and shared-expense use. (see the next table)

User Class	Description	Experience Level	Frequency of Use	Key Privileges
Guest	An unauthenticated visitor.	Low to moderate; may be new to FinPact.	One-time or occasional (registration/login) .	Register an account; log in. Cannot access any protected features.
Individual User (Registered)	An authenticated account holder who records and manages personal expenses, budgets, and analytics.	Low to moderate for financial tracking; expected to be comfortable with standard web forms.	Daily to weekly.	All personal-expense, budget, and analytics functions; settlement recording when participating in shared expenses.
Shared Expense Participant (Registered)	A registered user who participates in one or more shared expenses, views shares and balances, and records settlements.	Low to moderate.	Occasional, typically tied to specific shared bills (rent, dinners, trips).	View shared expenses in which they participate; track payments/debts; record settlements.

Training requirements: No Training/Tutorial Requirements for all user classes

Technical expertise:

- All user classes are expected to be comfortable with standard web forms and navigation.

- No user class is expected to have accounting, database, or programming knowledge.

2.4 Software Stakeholders (Market Launch Prospective)

The table below distinguishes **internal stakeholders** (the organization building/operating FinPact), **external stakeholders** (users, partners, regulators), and **indirect stakeholders** (affected by FinPact but not direct users). Each row notes whether the stakeholder is relevant to CSAI 203 Phase 2 or arises only at commercial launch.

2.4.1 External Stakeholders (User & Customer Side)

	Stakeholder	Point of Interest	Scope
E-01	Guest (prospective user)	Fast, low-friction registration; clear value proposition before sign-up.	CSAI 203(2)
E-02	Registered Individual User	Accurate personal expense tracking, clear budgets, understandable analytics.	CSAI 203(2)
E-03	Shared Expense Creator	Easy bill creation, fair splitting, control over participants and split method.	CSAI 203(2)
E-04	Shared Expense Participant	Transparency of their share and balances; ability to record settlement without disputes.	CSAI 203(2)
E-05	Paying Customer / Subscriber (if a freemium or paid tier exists)	Value for money, billing accuracy, cancellation, refunds.	Launch
E-06	Households / Roommates / Travel Groups. (collective user unit)	Group-level fairness; shared visibility of balances; dispute resolution.	Launch
E-07	Small Teams / Clubs / Student Groups (organizational users)	Bulk participant management, recurring shared expenses (e.g., club dues).	Launch
E-08	Accessibility Users (users relying on screen readers, keyboard-only navigation, high-contrast displays)	Compliance with C-11 and basic accessibility expectations. (Check the constraints subsection)	Launch
E-09	International / Multi-currency Users	Currency handling explicitly out of scope in Phase 2 (A-08) but a market expectation. (Check the assumptions Subsection)	Launch
E-10	Minors / Guardians (if FinPact is used by dependents)	Age-appropriate access, parental consent, data minimization for minors.	Launch

2.4.2. Internal Stakeholders (Organization Side):

	Stakeholder	Point of Interest	Scope
I-01	Product Owner / Founders	Product vision, scope, prioritization, ROI, competitive positioning against Splitwise, Tricount, and similar apps.	CSAI 203(2)
I-02	Product Management	Feature roadmap, requirement trade-offs, acceptance criteria, release scope.	CSAI 203(2)
I-03	Engineering Team	Feasibility, MVC design, maintainability, deployment,	CSAI 203(2)

	(Developers, Architects, DevOps)	performance, security.	
I-04	UX / UI Design	Usability of expense entry, split flows, budget dashboards, and mobile layout.	CSAI 203(2)
I-05	QA / Test Engineering	Test coverage against FR-01–FR-20 and NFR-01–NFR-05; regression safety.	CSAI 203(2)
I-06	Customer Support / Success	Handling user issues (e.g., “my balance is wrong”, “I can’t log in”), refund requests, FAQ content.	Launch
I-07	Marketing & Growth	Positioning, onboarding funnel, retention, referral loops (e.g., “invite your roommates”).	Launch
I-08	Finance / Accounting (of the company)	Subscription revenue, operating costs, tax obligations for the business itself.	Launch
I-09	Legal & Compliance	Terms of Service, Privacy Policy, data-processing agreements, consumer-protection compliance.	Launch
I-10	Security & Privacy Team	Password hashing, session security, breach response, data-retention policy (C-13), privacy review.	Launch
I-11	Investors / Board	Growth metrics, user acquisition cost, retention, revenue trajectory.	Launch

***Note:** Phase 2 scope**,** Only stakeholders marked **“CSAI 203(2)”** impose Phase 2 requirements and are traced to FRs/NFRs. Stakeholders marked **“Launch”** are recorded here for completeness and for the transition from course project to market product, but they do **not** generate Phase 2 requirements. This preserves the IEEE 830 rule that Section 2.4 describes stakeholders without silently expanding scope.

2.5 Operating Environment

FinPact operates in the following environment:

2.5.1 Client environment:

- Modern desktop and mobile web browsers: Google Chrome, Mozilla Firefox, Edge..etc

- Supported viewports: ≥ 360 × 640 px (mobile) and ≥ 1366 × 768 px (desktop).

- Requires JavaScript enabled and HTTPS access.

- No browser plug-ins or special fonts are required.

2.5.2 Server environment:

- A web application server supporting the course-approved runtime and MVC framework.

- A relational database management system (e.g., MySQL, PostgreSQL).

- HTTPS/TLS termination at the server or at the hosting platform.

- Availability of a secure password-hashing library.

2.5.3 Network environment:

- Internet connectivity between users and the FinPact server.

- Internal connectivity between the FinPact server and the database.

2.5.4 Operating system:

- FinPact is OS-agnostic on the client side (browser-based).

- The server OS is dictated by the hosting environment and the course-approved stack.

2.5.5 Development and test environment:

- Local development using the course-approved stack (C-01) (check the constraints section) and an MVC framework.

- Version control (e.g., Git) and a shared repository for the project team.

2.5.6 Expected data scale:

- Course-project scale: up to roughly 100 registered users

- NFR-01 defines the formal performance target for this scale.

2.6 Design and Implementation Constraints

The following constraints limit how FinPact may be designed and built. Where a constraint is verifiable, the related requirement is listed.

ID	Constraint	Description	Related
C-01	Course Technology Stack	FinPact shall be implemented using only the backend and frontend frameworks requested by the instructor.	Course rules 17, 19
C-02	MVC Architecture	The system shall follow the Model View Controller pattern. Models hold entities and business rules (split calculation, balances, budget usage); Views render pages and contain no business logic; Controllers receive requests, validate input, call Models, and select Views. MVC shall be visible in both the Design Document and the /src structure.	Course rule 21
C-03	Web-Based Delivery	FinPact is delivered only as a web application; no native mobile or desktop client is built.	1.2.2

C-04	No Real Financial Transactions	FinPact shall not transfer money, connect to payment gateways or banks, or store card/bank data. Payments and settlements are informational records only.	FR-12, FR-20
C-05	Server-Side Validation	Every input is validated on the server, even when the browser also validates it.	NFR-04
C-06	Session-Based Authentication	All features except registration and login require an authenticated session, which is invalidated on logout or after inactivity.	FR-02, NFR-02
C-07	Responsive UI	The UI shall remain usable on the mobile and desktop viewports listed in 2.5.1.	NFR-05
C-08	Single UI Language	The UI is presented in English only; localization is out of scope.	—
C-9	Data Minimization	Only name, email, password hash, and the financial records required by FR-01–FR-20 are collected. No phone number, national ID, address, or card/bank data is requested.	FR-01
C-10	Financial Data Confidentiality	A user's personal expenses, budgets, and notifications are visible only to that user; a shared expense is visible only to its creator and participants (A-13). Checks are enforced on the server.	NFR-02
C-11	Data Retention	Records are kept for the lifetime of the account unless the user deletes them; no automatic archiving or purging is performed.	FR-05
C-12	Error Handling	User-facing error messages shall not reveal stack traces, SQL, file paths, or other internal details.	NFR-02

2.7 User Documentation

The following user-facing documentation is delivered digitally (PDF and/or in-application pages) with the application:

User Guide (Video): registration and login, personal expenses, shared expenses and splitting, payments and settlements, budgets and category limits, and analytics.

FAQ: common questions, e.g., how an equal split is rounded, what a pending settlement means, and why a budget warning appeared.

In-Application Help: short help text and tooltips on the split, settlement, and budget pages explaining terms such as share, balance, threshold, and settlement.

2.8 Assumptions and Dependencies

A-01	Users have reliable internet access and a supported browser.

A-02	Users enter accurate expense, payment, budget, and settlement data; FinPact does not verify financial accuracy against any external source.
A-03	Each email address identifies exactly one FinPact account.
A-04	Budget periods are calendar months; custom period lengths are not supported.
A-05	Every shared expense has at least two participants, including the creator
A-06	Shares are split equally or by custom amounts that sum to the total; percentage and weighted splits are not supported
A-07	Settlements are informational records only; FinPact does not verify that a real-world transfer took place.
A-08	Each account uses a single currency (EGP by default); multi-currency conversion is not supported.
A-09	No external compliance regime (e.g.GDPR) is imposed beyond institutional computing-use policy, because no card or bank data is processed
A-10	Expense categories are a fixed, predefined list (Food, Transportation, Shopping, Bills, Entertainment, Education, Other); user-defined categories are out of scope
A-11	A shared expense is visible only to its creator and its participants.

2.9 Apportioning of Requirements

All functional requirements FR-01–FR-20 and non-functional requirements NFR-01–NFR-05 are in scope. Implementation is apportioned across the course milestones as follows, so that the Core Functionality Prototype delivers about 50% of the functionality and the Final Delivery delivers 100%:

Milestone	Requirements Delivered	Rationale
Core Functionality Prototype (Phase 4)	FR-01–FR-07, FR-09–FR-12 (11 of 20 FRs); NFR-02 and NFR-04 for these features	Accounts, personal expenses, and the shared-expense core are prerequisites for every other feature.
Final Delivery (Phase 5)	FR-08, FR-13–FR-20; NFR-01, NFR-03, NFR-05 verified on the complete system; deployment	Budgets, analytics, and settlements depend on stable expense and balance data.

Note: The following capabilities are out of scope and are recorded only as candidates for future versions: email or push notifications, recurring expenses, percentage or weighted splits, multi-currency support, receipt image upload, CSV/PDF export, native mobile applications, payment-gateway integration, and third-party API access. They are not traced to any FR or use case and are excluded from acceptance testing.

3. Requirements Specification

This section states FinPact's requirements in enough detail to design, implement, and test the system. Section 3.1 specifies the functional requirements, 3.2 the use cases, 3.3 the conceptual class model, 3.4 the non-functional requirements, and 3.5 the external interfaces. The keywords "shall", "should", and "may" follow Section 1.6.

3.1 Functional Requirements

Each requirement below lists its inputs, processing rules, outputs, priority, and traceability to use cases (Section 3.2) and classes (Section 3.3). Rules that apply to every requirement are stated once in Section 3.1.1 and are not repeated.

3.1.1 Common Rules

ID	Rule (applies to all FRs unless stated otherwise)
CR-1	**Authentication.** Every function except registration and login requires an authenticated session. Unauthenticated requests are redirected to the login page.
CR-2	**Ownership.** A user can view or change only their own personal expenses, budgets, limits, and notifications, and only shared expenses they created or participate in. Any other request is rejected with "Access denied" and changes nothing.
CR-3	**Amounts.** Monetary amounts are positive decimals with at most two decimal places and a maximum of 1,000,000.00, in the account currency (A-08).
CR-4	**Feedback and integrity.** A successful create, update, or delete shows a success message. A failed operation shows an error message without internal details (C-14) and leaves stored data unchanged (NFR-04).

3.1.2 Functional Requirements

User Account Management

FR-01: User Registration

Description	The system shall allow a guest to create a FinPact account.
Inputs	Name, email address, password, password confirmation.

Processing & Rules	1. All fields are required. 2. Name is 2–50 characters. 3. Email must have a valid format and must not already be registered (compared case-insensitively); it is stored in lowercase. 4. Password is at least 8 characters with at least one letter and one digit, and must match the confirmation.
Outputs	Account created; success message; redirect to the login page. If the email is taken, the message “This email is already registered” is shown and no account is created.
Priority	High
Traceability	UC-01 · User

FR-02: Login and Logout

Description	The system shall authenticate registered users and allow them to end their session.
Inputs	Email and password (login); Log Out action (logout).
Processing & Rules	1. The password is verified against the stored hash. 2. On failure, one generic message (“Invalid email or password”) is shown, without revealing which field was wrong. 3. On success, a new session identifier is issued and the user is redirected to the Dashboard. 4. A session expires after 30 minutes of inactivity. 5. Logout invalidates the session on the server; later requests using the old session are treated as unauthenticated.
Outputs	Dashboard after login; login page with a “You have been logged out” message after logout.
Priority	High
Traceability	UC-02, UC-03 · User

Personal Expense Management

FR-03: Add Personal Expense

Description	The system shall allow a user to record a personal expense.
Inputs	Amount, date (defaults to today), category (FR-07), optional description.
Processing & Rules	1. Amount, date, and category are required.. 2. The expense is linked to the authenticated user. 3. After saving, budget usage for the expense’s month is recalculated (FR-16) and threshold checks run (FR-17).
Outputs	Confirmation message; the expense appears in the expense history (FR-06).
Priority	High
Traceability	UC-04 · PersonalExpense, ExpenseCategory, User

FR-04: Edit Personal Expense

Description	The system shall allow a user to update the amount, date, category, or description of their own expense.
Inputs	Selected expense; updated field values.
Processing & Rules	1. Only the owner may edit. 2. The same validation as FR-03 applies; if any field is invalid, no field is saved. 3. Budget usage is recalculated for the affected month(s), both the old and the new month if the date changed, and FR-17 checks run.
Outputs	Updated expense shown with a confirmation message.
Priority	High
Traceability	UC-05 · PersonalExpense

FR-05: Delete Personal Expense

Description	The system shall allow a user to delete their own expense.
Inputs	Selected expense; confirmation.
Processing & Rules	1. Only the owner may delete. 2. A confirmation dialog is shown; cancelling leaves the expense unchanged. 3. On confirmation the expense is permanently removed, and budget usage and analytics are recalculated. 4. Budget notifications created earlier are kept as history.
Outputs	Expense removed from the history; confirmation message.
Priority	High
Traceability	UC-06 · PersonalExpense

FR-06: View Expense History

Description	The system shall display the user’s personal expenses.
Inputs	Page number (optional).
Processing & Rules	1. Only the authenticated user’s expenses are listed. 2. Each row shows amount, date, category, and description, sorted by date (newest first). 3. The list is paginated at 20 expenses per page, with the total amount of the listed expenses. 4. Each row offers Edit (FR-04) and Delete (FR-05).
Outputs	Expense list, or “No expenses recorded yet” with a link to Add Expense.
Priority	High
Traceability	UC-21 · PersonalExpense, ExpenseCategory

FR-07: Categorize Expenses

Description	The system shall require every personal expense to belong to exactly one category.
Inputs	Category selected from a drop-down list when adding or editing an expense.
Processing & Rules	1. Categories are the fixed list in A-16. 2. A value outside the list is rejected. 3. Changing an expense's category recalculates the affected category usage (FR-16).
Outputs	Expense stored with its category; category shown in the history, filters, budgets, and analytics.
Priority	High
Traceability	UC-07 · ExpenseCategory, PersonalExpense

FR-08: Search and Filter Expenses

Description	The system shall allow a user to find expenses by keyword and filter criteria.
Inputs	Keyword; one or more categories; date range (from–to); amount range (min–max). All optional.
Processing & Rules	1. The keyword is matched case-insensitively against the description. 2. All provided criteria are combined (AND); empty criteria are ignored. 3. "From" must not be after "To", and "Min" must not exceed "Max"; otherwise the search is not run. 4. Results contain only the user's own expenses, sorted as in FR-06; stored data is not modified. 5. "Clear filters" restores the full history.
Outputs	Matching expenses with their count and total amount, or "No results found".
Priority	Medium
Traceability	UC-08 (extends UC-21) · PersonalExpense

Shared Expense Management

FR-09: Create Shared Expense

Description	The system shall allow a user to record a bill shared with other registered users.
Inputs	Title, total amount, date, optional description, participants (FR-10), split method and shares (FR-11), and optionally the participant who paid the bill.

Processing & Rules	1. CR-3, CR-4, and CR-5 apply; the creator is added as a participant automatically. 2. The expense, its participants, and its shares are saved together; if any part fails, nothing is saved (NFR-04). 3. If a payer is given, a payment record for the full amount is created for that participant (FR-12). 4. Only the creator may change the title, total, or participants, and only until the first settlement of the expense is confirmed; any change to the total or participants requires a new split (FR-11).
Outputs	Shared expense details page showing participants and shares; the expense appears in every participant's shared history (FR-13).
Priority	High
Traceability	UC-09 · SharedExpense, SharedExpenseParticipant, User

FR-10: Add Participants

Description	The system shall allow the creator to add registered users as participants of a shared expense.
Inputs	Email address of each participant.
Processing & Rules	1. A participant is found by exact email match; only the matched user's name is shown, and the full user list is never exposed (C-12). 2. Only registered users can be added; the same user cannot be added twice. 3. A shared expense has 2–20 participants, including the creator. 4. Participants may be removed before the split is confirmed; the minimum of two is re-checked.
Outputs	Updated participant list.
Priority	High
Traceability	UC-10 · SharedExpenseParticipant, User

FR-11: Split Shared Bill

Description	The system shall calculate each participant's share of a shared expense.
Inputs	Split method (Equal or Custom); for Custom, an amount per participant.
Processing & Rules	1. **Equal:** each share is the total divided by the number of participants, rounded down to two decimals; the remaining piasters are added to the creator's share. Example: EGP 100.00 for 3 people → 33.33, 33.33, and 33.34 (creator). 2. **Custom:** each share is at least 0.01, and the shares must add up exactly to the total; otherwise the split is rejected and the difference is shown. 3. The share of every participant is stored; the sum of shares always equals the total (NFR-04). 4. Shares are recalculated whenever the total or the participants change (FR-09 rule 4).
Outputs	Each participant's share displayed for confirmation, then saved.
Priority	High

Traceability	UC-11 · SharedExpenseParticipant

FR-12: Track Payments and Debts

Description	The system shall record who paid towards a shared expense and show how much each participant owes or is owed.
Inputs	Payer (a participant), amount, payment date, optional description.
Processing & Rules	1. Only the creator or a participant may record or view payments; the payer must be a participant. 2. The total of all payments of an expense cannot exceed its total amount. 3. Net balance of a participant = amount paid − share + confirmed settlements sent − confirmed settlements received. Positive means the participant is owed money; negative means they owe money. 4. A “who owes whom” list matches debtors to creditors, largest amounts first. 5. Status is **Open** while any balance is non-zero or payments are below the total, and **Settled** when payments equal the total and every balance is zero. Example: one person pays EGP 1,200 for four people (EGP 300 each); their balance is +900 and each of the other three owes them EGP 300.
Outputs	Per participant: share, amount paid, and net balance; the “who owes whom” list; the expense status. The Dashboard shows the user’s totals “You owe” and “You are owed” across all shared expenses.
Priority	High
Traceability	UC-12 · PaymentRecord, SharedExpenseParticipant, Settlement

FR-13: View Shared Expense History

Description	The system shall display the shared expenses the user created or participates in.
Inputs	Optional status filter (Open / Settled).
Processing & Rules	1. Only shared expenses where the user is creator or participant are listed. 2. Each row shows title, date, total, the user’s share, the user’s net balance, and status, newest first. 3. Selecting a row opens the details: participants and shares, payments, settlements with their status, and the “who owes whom” list.
Outputs	Shared expense list and details, or “No shared expenses yet” with a link to Create Shared Expense.
Priority	High
Traceability	UC-13 · SharedExpense, SharedExpenseParticipant, PaymentRecord, Settlement

Budget Management

FR-14: Set Monthly Budget

Description	The system shall allow a user to set a total spending budget for a calendar month.
Inputs	Month (YYYY-MM, defaults to the current month) and total amount.
Processing & Rules	1. A user has at most one budget per month; setting a budget for a month that already has one updates it after confirmation. 2. Only the current or a future month can be created or changed; past months are read-only. 3. The total cannot be lower than the sum of that month's category limits (FR-15). 4. After saving, usage is recalculated (FR-16) and threshold checks run (FR-17).
Outputs	Budget saved; usage progress bar displayed.
Priority	High
Traceability	UC-15 · Budget

FR-15: Set Category Spending Limits

Description	The system shall allow a user to set a spending limit for a category within a monthly budget.
Inputs	Month, category, limit amount.
Processing & Rules	1. A monthly budget must exist for the selected month. 2. Each category has at most one limit per budget; setting it again replaces the old value. 3. The sum of all category limits must not exceed the monthly total; otherwise the limit is rejected and the remaining available amount is shown. 4. A limit may be removed; categories without a limit count only towards the monthly total.
Outputs	Category limit saved; category progress bar displayed.
Priority	High
Traceability	UC-16 · CategoryBudgetLimit, Budget, ExpenseCategory

FR-16: Monitor Budget Usage

Description	The system shall display the spent amount, remaining amount, and usage percentage of the monthly budget and of each category limit.
Inputs	Month (defaults to the current month).
Processing & Rules	1. Spent = sum of the user's personal expenses dated in that month; for a category, only expenses of that category. 2. Remaining = budget − spent; a negative value is shown as "Over budget by X". 3. Usage % = spent ÷ budget × 100, rounded to one decimal place. 4. Progress bars use three states: Normal (< 80%), Warning (80–99.9%), Exceeded (≥ 100%). 5. Values are recalculated after every add, edit, or delete of an expense and every budget or limit change.

Outputs	Budget overview on the Budgets page and a summary on the Dashboard; “No budget set for this month” with a link to FR-14 if none exists.
Priority	High
Traceability	UC-17 · Budget, CategoryBudgetLimit, PersonalExpense

FR-17: Budget Limit Warning

Description	The system shall warn the user when spending reaches 80% or 100% of the monthly budget or of a category limit.
Inputs	Triggered by FR-03, FR-04, FR-14, or FR-15; “Mark as read” action.
Processing & Rules	1. When usage reaches a threshold, a budget notification is stored with message, threshold (80 or 100), creation time, and isRead = false, linked to the budget and, for a category, to its limit. 2. At most one notification is created per threshold per budget or limit; if usage jumps directly past 100%, only the 100% notification is created. 3. The warning appears as a banner right after the triggering action and stays in the notification list. 4. Notifications are shown in the application only; email or push notifications are out of scope. Example: “You have used 82.5% of your Food limit for October 2026 (EGP 1,650 of 2,000).”
Outputs	Warning banner; notification list with an unread count on the Dashboard; marking as read sets isRead = true.
Priority	High
Traceability	UC-18 · BudgetNotification, Budget, CategoryBudgetLimit

Spending Analytics

FR-18: View Spending Analytics

Description	The system shall summarize the user’s spending by category for a selected period.
Inputs	A month (default: current) or a custom date range of up to 12 months.
Processing & Rules	1. Calculated from personal expenses only. 2. Shows the period total and, for each category, its total, percentage of the period total, and number of expenses, sorted by amount (highest first). 3. Categories with no spending are omitted. 4. Results are shown as a chart and a table; no data is modified.
Outputs	Category summary chart and table, or “No expenses in this period”.
Priority	Medium
Traceability	UC-19 · PersonalExpense, ExpenseCategory

FR-19: View Spending Trends

Description	The system shall allow a user to compare spending across months.
Inputs	Two months to compare (default: current and previous month).
Processing & Rules	1. For each category and in total: amount in each month, difference, and change % = (B − A) ÷ A × 100; when A is zero the change is shown as “New”. 2. A chart shows monthly totals for the last six months. 3. The same month cannot be selected twice.
Outputs	Comparison table and six-month trend chart.
Priority	Medium
Traceability	UC-20 · PersonalExpense, ExpenseCategory

Settlement Management

FR-20 — Settle Outstanding Amounts

Description	The system shall allow participants to record settlements of outstanding amounts and update balances accordingly.
Inputs	Shared expense, receiver, amount, optional note; Confirm or Reject action by the receiver.
Processing & Rules	1. A settlement is recorded by the debtor (a participant with a negative balance) towards a creditor (a participant with a positive balance) in the same expense. 2. The amount must be greater than zero and not exceed the debtor’s debt or the receiver’s credit; partial settlements are allowed. 3. The settlement is saved with status **Pending** and settlementDate set by the server. 4. The receiver confirms (status **Confirmed**) or rejects (status **Rejected**). Only confirmed settlements change balances (FR-12); rejected ones remain as history. 5. When all balances reach zero, the expense becomes Settled. 6. No money is moved; the form states this clearly.
Outputs	Settlement listed on the expense with its status; updated balances after confirmation.
Priority	High
Traceability	UC-14 · Settlement, SharedExpense, User

3.2 Use Case Model

The use-case model is presented as three diagrams, one per group of related functions, followed by the descriptions of all 21 use cases. Each use case traces to the functional requirement it realizes (Section 3.1).

3.2.1 Use Case Diagrams

«include» marks behaviour that is always part of the base use case; «extend» marks optional behaviour that is added at an extension point when its condition holds. Actors outside the system boundary are the Guest, the Registered User, and, for settlements, the Receiving Participant (a registered user in a secondary role).

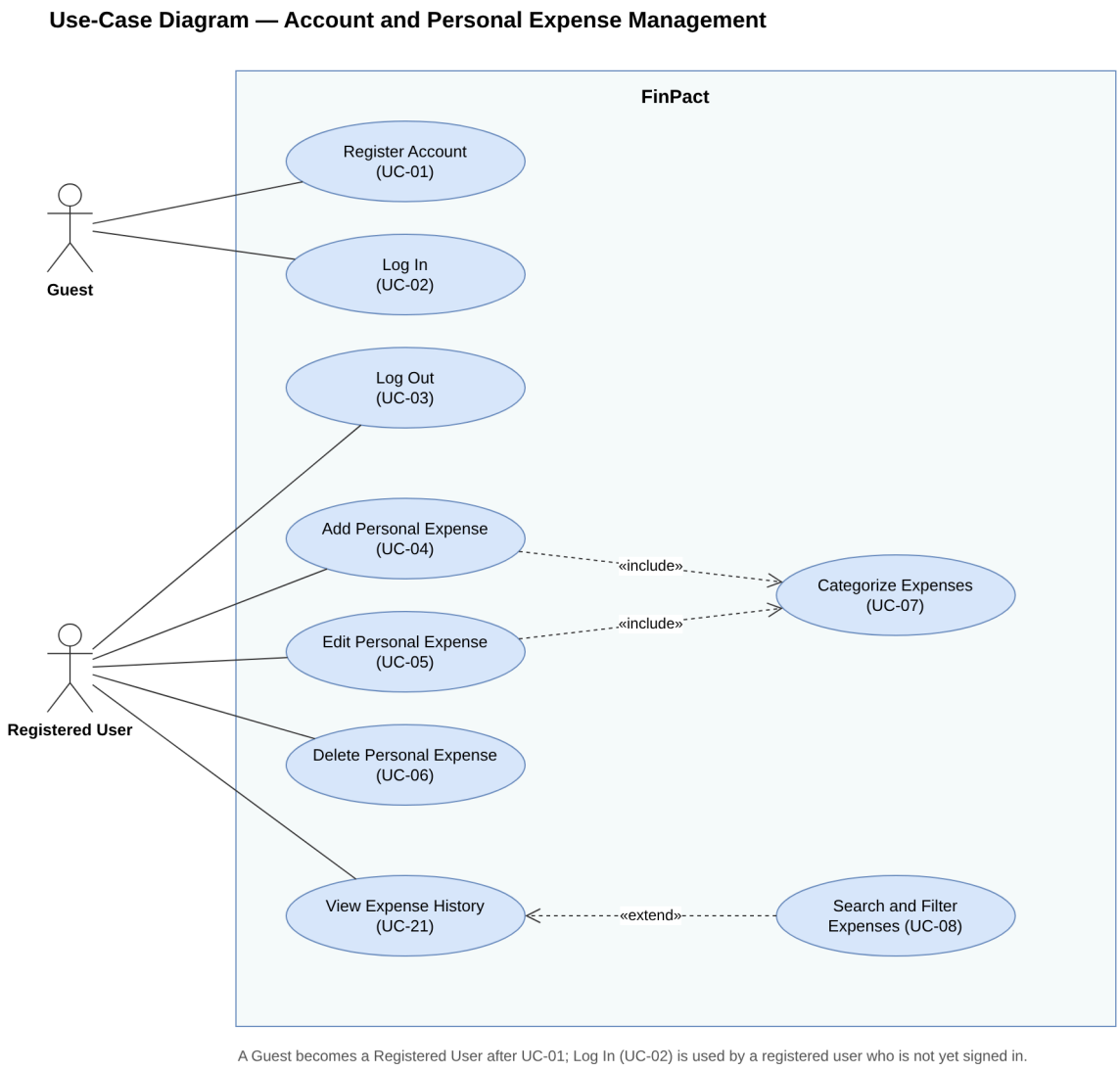


Figure 2: Use-Case Diagram : Account and Personal Expense Management (made via Draw.io)

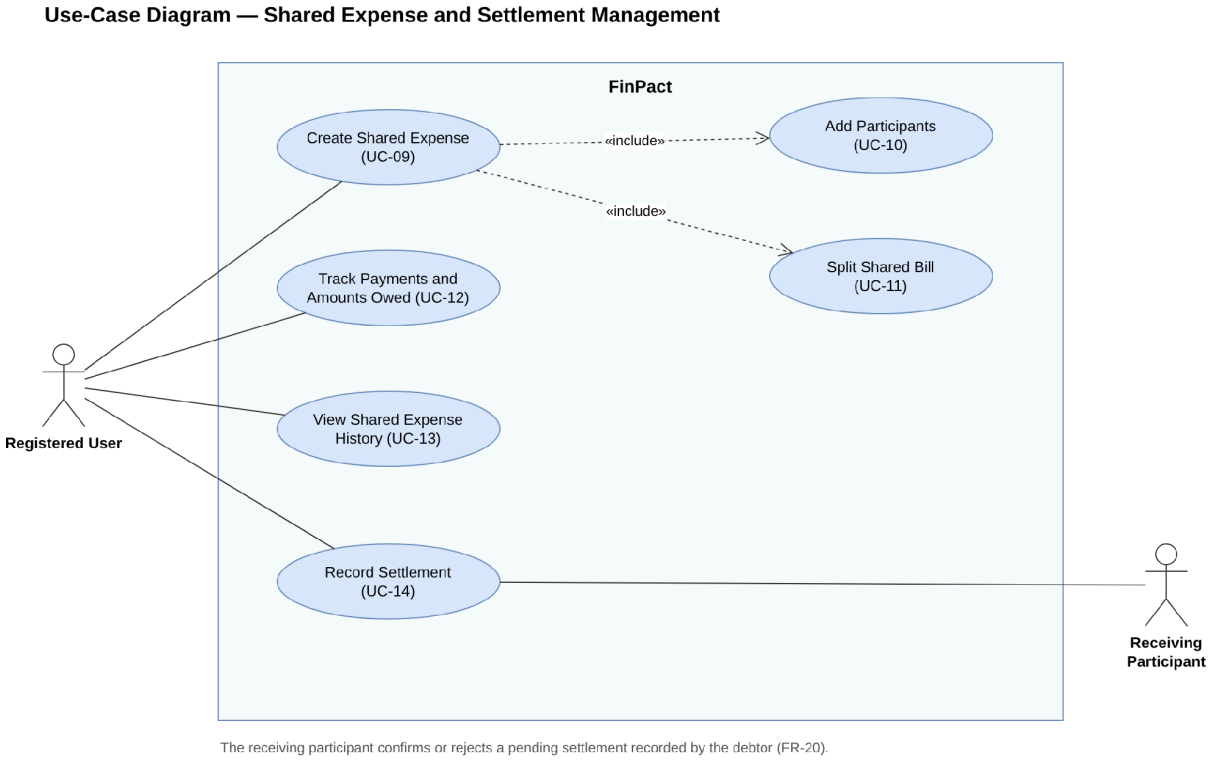


Figure 3: Use-Case Diagram : Shared Expense and Settlement Management

Use-Case Diagram — Budget Management and Spending Analytics

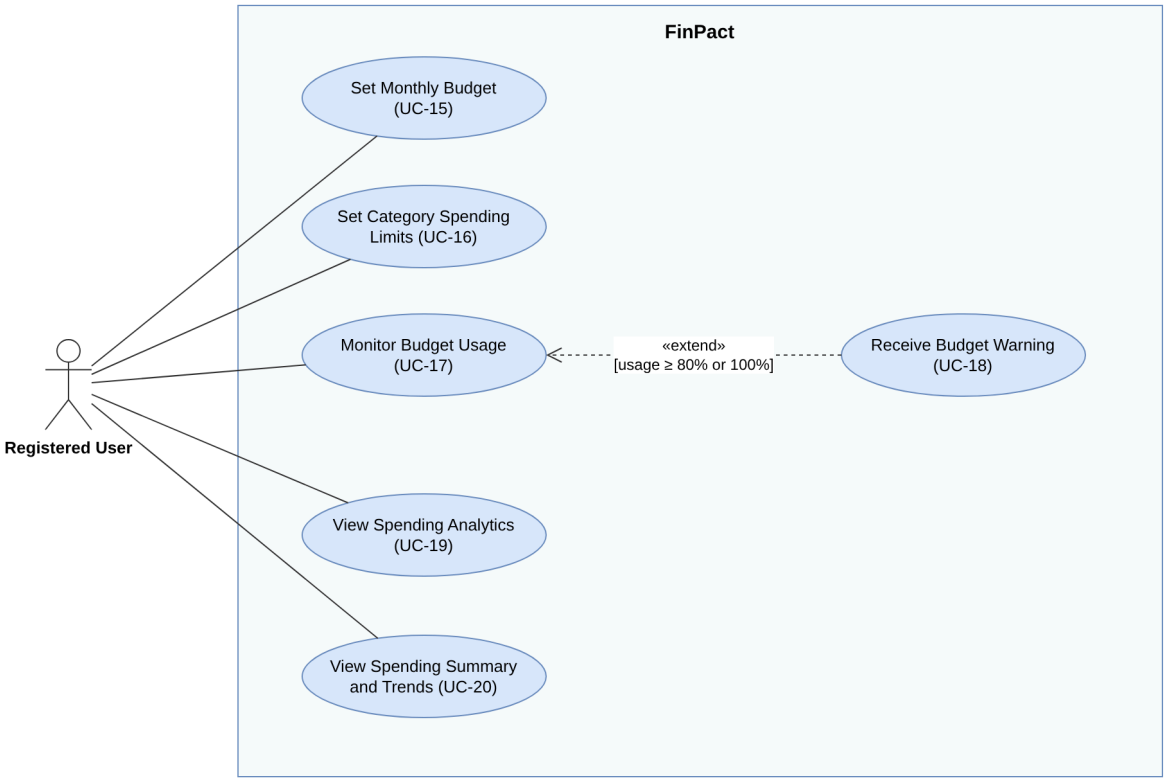


Figure 4: Use-Case Diagram: Budget Management and Spending Analytics

3.2.2 Use-Case Descriptions

All use cases follow the same IEEE-based template (reference [5]). Rules shared by every use case, authentication, ownership, amount, date, and text validation, and error handling are stated once in Section 3.1.1 and are not repeated in each description.

UC-01: Register Account

Use Case ID	UC-01
Use Case Name	Register Account
Related FR	FR-01
Primary Actor	Guest
Secondary Actors	None
Description	This case describes how a new user creates an account in FinPact to access the application's personal expense management, shared expense tracking, and budgeting features.
Trigger	The guest selects the Register option to create a new FinPact account.
Preconditions	1. The guest is not signed in to an existing FinPact account. 2. The guest can access the registration page.
Postconditions	1. A new user account is created successfully. 2. The account is available for subsequent login.
Normal Flow	1. The guest opens the FinPact registration page. 2. The system displays the registration form. 3. The guest enters the required information: name, email address, password, and password confirmation. 4. The system validates the submitted information. 5. The system checks whether the email address is already registered. 6. The system creates the account if the information is valid and the email address is available. 7. The system displays a success message confirming that the account has been created. 8. The system redirects the guest to the login page.
Alternative Flows	A1. Already Registered: 1. At step 2 the guest selects the “Log in instead” link. 2. The system opens the login page (UC-02); no account is created.
Exceptions	E1. Missing Required Information: The system identifies the missing information and asks the guest to complete the required fields. E2. Email Already Registered: The system displays a message indicating that the email address is already in use and does not create a duplicate account. E3. Invalid Registration Information: The system displays an appropriate validation message and asks the guest to correct the invalid information. E4. Account Creation Failure: If the account cannot be created, the system displays an error message and does not indicate that registration was successful.

Includes / Extends	None
Priority	High
Frequency of Use	Once per user.
Business Rules	FR-01 rules 1–5: name 2–50 characters; unique, case-insensitive email stored in lowercase (A-03); password of at least 8 characters with a letter and a digit, matching its confirmation, stored only as a salted hash (C-04). Only name, email, and password are collected (C-10).

UC-02: Log In

Use Case ID	UC-02
Use Case Name	Log In
Related FR	FR-02
Primary Actor	Registered User
Secondary Actors	None
Description	This case describes how a registered user logs in to FinPact to access the application's expense management and budgeting features.
Trigger	The registered user selects the Log In option.
Preconditions	1. The user has an existing FinPact account. 2. The user is not currently signed in. 3. The user can access the login page.
Postconditions	1. The user is successfully authenticated. 2. The user gains access to the features available to their account.
Normal Flow	1. The user opens the FinPact login page. 2. The system displays the login form. 3. The user enters their registered email address and password. 4. The user submits the login form. 5. The system validates the submitted information. 6. The system verifies the credentials against the registered account. 7. The system authenticates the user. 8. The system displays the Dashboard.
Alternative Flows	A1. Redirect to Requested Page: 1. The user was sent to the login page while trying to open a protected page. 2. After step 7 the system opens the originally requested page instead of the Dashboard.

Exceptions	E1. Missing Login Information: The system identifies the missing email address or password and asks the user to complete the required fields. E2. Invalid Credentials: If the email address or password is incorrect, the system displays the generic message “Invalid email or password”, without revealing which field was wrong, and does not grant access. E3. Login Failure: If the system cannot complete the login process, it displays an appropriate error message and does not indicate that login was successful.
Includes / Extends	None
Priority	High
Frequency of Use	Several times per week per user.
Business Rules	One generic message for any failed login (FR-02 rule 2); a new session identifier is issued at login; sessions expire after 30 minutes of inactivity (C-07).

UC-03: Log Out

Use Case ID	UC-03
Use Case Name	Log Out
Related FR	FR-02
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a signed-in user logs out of FinPact to end their current session.
Trigger	The user selects the Log Out option.
Preconditions	1. The user is signed in to FinPact. 2. The user can access the Log Out option.
Postconditions	1. The user's current session is terminated. 2. The user is no longer authenticated and must log in again to access account features.
Normal Flow	1. The user selects the Log Out option. 2. The system receives the logout request. 3. The system terminates the user's current session. 4. The system redirects the user to the login page. 5. The system confirms that the user has been logged out.
Alternative Flows	A1. Session Timeout: 1. The user is inactive for 30 minutes. 2. The system ends the session automatically. 3. The next request is redirected to the login page with the message “Your session has expired”.

Exceptions	E1. Logout Processing Failure: If the system cannot complete the logout process, it displays an appropriate error message and does not confirm that logout was successful.
Includes / Extends	None
Priority	High
Frequency of Use	Once per session.
Business Rules	Logout invalidates the session on the server; requests that reuse the old session are treated as unauthenticated (FR-02 rule 5, C-07).

UC-04: Add Personal Expense

Use Case ID	UC-04
Use Case Name	Add Personal Expense
Related FR	FR-03, FR-07
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a registered user records a new personal expense in FinPact to maintain an accurate record of personal spending.
Trigger	The user selects the Add Expense option.
Preconditions	1. The user is signed in to FinPact. 2. The user can access the personal expense management feature.
Postconditions	1. The new personal expense is saved successfully. 2. The expense appears in the user's personal expense records. 3. Budget usage for the expense’s month is recalculated.
Normal Flow	1. The user selects the Add Expense option. 2. The system displays the expense entry form. 3. The user enters the required expense information, including the amount, date (default: today), category (UC-07), and an optional description. 4. The user submits the expense information. 5. The system validates the submitted information. 6. The system saves the expense, associates it with the user's account, and recalculates the budget usage of its month (UC-17). 7. The system displays a confirmation message indicating that the expense was added successfully.
Alternative Flows	A1. Cancel: 1. Before step 4 the user selects Cancel. 2. The system discards the entered data and returns to the expense history; nothing is saved.

Exceptions	E1. Missing Required Information: The system identifies any missing required fields and asks the user to complete them. E2. Invalid Expense Information: If the submitted information is invalid, the system displays an appropriate validation message and asks the user to correct the information. E3. Expense Saving Failure: If the system cannot save the expense, it displays an error message and does not indicate that the operation was successful.
Includes / Extends	«include» Categorize Expenses (UC-07)
Priority	High
Frequency of Use	Several times per day per active user.
Business Rules	CR-1–CR-6. Amount, date, and category are required; the date defaults to today. Saving triggers budget recalculation (FR-16) and threshold checks (FR-17).

UC-05: Edit Personal Expense

Use Case ID	UC-05
Use Case Name	Edit Personal Expense
Related FR	FR-04, FR-07
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a registered user modifies an existing personal expense in FinPact to keep their expense records accurate.
Trigger	The user selects an existing expense from their personal expense history and chooses the Edit option.
Preconditions	1. The user is signed in to FinPact. 2. The selected personal expense exists and belongs to the authenticated user.
Postconditions	1. The expense is updated successfully with the new information. 2. Any affected expense totals, budget usage, or analytics are recalculated as appropriate.
Normal Flow	1. The user opens their personal expense history. 2. The user selects the expense they want to modify. 3. The user chooses the Edit option. 4. The system displays the current expense information. 5. The user updates one or more fields, such as the amount, category (UC-07), date, or description. 6. The user submits the changes. 7. The system validates the updated information. 8. The system saves the changes. 9. The system displays the updated expense and confirms that the changes were saved successfully.

Alternative Flows	A1. Cancel Edit: 1. Before step 6 the user selects Cancel. 2. The system discards the changes and shows the expense unchanged.
Exceptions	E1. Expense No Longer Exists: If the selected expense no longer exists, the system informs the user and does not perform the update. E2. Unauthorized Expense Access: If the selected expense does not belong to the authenticated user, the system denies the edit and does not modify the expense. E3. Invalid Updated Information: If any updated information is invalid, the system displays an appropriate message and asks the user to correct it. Invalid changes are not saved. E4. Saving Failure: If the system cannot save the changes, it displays an error message and preserves the last successfully stored version of the expense.
Includes / Extends	«include» Categorize Expenses (UC-07)
Priority	High
Frequency of Use	A few times per week.
Business Rules	Only the owner may edit (CR-2). Validation is all-or-nothing. Budget usage is recalculated for both the old and the new month if the date changes (FR-04).

UC-06: Delete Personal Expense

Use Case ID	UC-06
Use Case Name	Delete Personal Expense
Related FR	FR-05
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a registered user removes an existing personal expense from their expense records in FinPact.
Trigger	The user selects an existing personal expense and chooses the Delete option.
Preconditions	1. The user is signed in to FinPact. 2. The selected personal expense exists and belongs to the authenticated user.
Postconditions	1. The expense is no longer included among the user's active personal expenses. 2. Any affected expense totals, budget usage, or analytics are updated as appropriate.

Normal Flow	1. The user opens their personal expense history. 2. The user selects the expense they want to delete. 3. The user chooses the Delete option. 4. The system requests confirmation of the deletion. 5. The user confirms the deletion. 6. The system removes the expense from the user's active expense records permanently (C-13). 7. The system updates the expense history and any affected calculations. 8. The system displays a confirmation message indicating that the expense was deleted successfully.
Alternative Flows	A1. Deletion Cancelled: 1. At step 5 the user chooses to cancel the deletion. 2. The system cancels the operation. 3. The expense remains unchanged.
Exceptions	E1. Expense Does Not Exist: If the selected expense no longer exists, the system informs the user and does not perform the deletion. E2. Unauthorized Expense Access: If the selected expense does not belong to the authenticated user, the system denies the deletion and leaves the expense unchanged. E3. Deletion Failure: If the system cannot complete the deletion, it displays an error message and does not indicate that the operation was successful.
Includes / Extends	None
Priority	High
Frequency of Use	A few times per month.
Business Rules	Only the owner may delete (CR-2). Deletion is permanent (C-13); budget notifications created earlier are kept as history (FR-05).

UC-07: Categorize Expenses

Use Case ID	UC-07
Use Case Name	Categorize Expenses
Related FR	FR-07
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a registered user assigns a category to a personal expense, either while adding it or while editing it, so that expenses can be organized, filtered, tracked against category limits, and analyzed.
Trigger	The user selects a category field while adding or editing a personal expense.
Preconditions	1. The user is signed in to FinPact. 2. The user is on the Add Expense or Edit Expense form. 3. At least one expense category is available in the system.

Postconditions	1. The selected category is stored with the expense. 2. The expense is included in category-based filtering, budget tracking, and spending analytics.
Normal Flow	1. The user opens the Add Expense or Edit Expense form. 2. The system displays the list of available categories (e.g., Food, Transportation, Shopping, Bills, Entertainment, Education, Other). 3. The user selects one category. 4. The user submits the expense form. 5. The system verifies that the selected category is valid and available. 6. The system stores the category association with the expense. 7. The system displays the expense under its selected category.
Alternative Flows	A1. Change Category: 1. The user edits an existing expense and selects a different category. 2. The system validates the new category and replaces the old association. 3. The system recalculates any affected budget usage and analytics.
Exceptions	E1. No Category Selected: The system asks the user to select a category and does not save the expense. E2. Invalid or Unavailable Category: The system rejects the value and asks the user to choose a valid category. E3. Saving Failure: The system displays an error message and does not indicate that the category was saved.
Includes / Extends	Included by UC-04 and UC-05
Priority	High
Frequency of Use	With every expense that is added or edited.
Business Rules	Exactly one category per expense, taken from the fixed list in A-16 (FR-07).

UC-08: Search and Filter Expenses

Use Case ID	UC-08
Use Case Name	Search and Filter Expenses
Related FR	FR-08
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a registered user finds specific personal expenses in their history by entering a search term or applying filters such as category, date range, and amount threshold. It extends the View Expense History use case.
Trigger	The user enters a search term or selects filter criteria on the expense history page.
Preconditions	1. The user is signed in to FinPact. 2. The user is on the expense history page.

Postconditions	1. The system displays only the user's own expenses that match the criteria. 2. The stored expense data is not modified.
Normal Flow	1. The user opens the expense history page. 2. The user enters a search term and/or selects one or more filters (category, date range, minimum/maximum amount). 3. The user submits the search. 4. The system validates the criteria. 5. The system retrieves the authenticated user's expenses that match all the selected criteria. 6. The system displays the matching expenses with their amount, category, date, and description.
Alternative Flows	A1. Clear Filters: 1. The user selects the Clear Filters option. 2. The system removes all criteria and displays the full expense history.
Exceptions	E1. No Matching Results: The system displays a clear "no results found" message. E2. Invalid Criteria: If the date range is invalid (e.g., start date after end date) or the amount is not a valid number, the system asks the user to correct it and does not run the search. E3. Retrieval Failure: The system displays an error message instead of misleading or empty results. E4. Unauthorized Data: The system never returns expenses that belong to another user.
Includes / Extends	«extend» View Expense History (UC-21) at the extension point "search criteria entered"
Priority	Medium
Frequency of Use	Weekly.
Business Rules	Criteria are combined with AND; "From" ≤ "To" and "Min" ≤ "Max"; results include only the user's own expenses (FR-08, CR-2).

UC-09: Create Shared Expense

Use Case ID	UC-09
Use Case Name	Create Shared Expense
Related FR	FR-09
Primary Actor	Registered User (creator)
Secondary Actors	None
Description	This use case describes how a registered user records a bill or expense shared with other people (e.g., rent, electricity, a group dinner). The use case includes Add Participants and Split Shared Bill.
Trigger	The user selects the Create Shared Expense option.

Preconditions	1. The user is signed in to FinPact. 2. The user can access the shared expense feature.
Postconditions	1. The shared expense is saved with the creator recorded as its owner. 2. The expense has a valid participant list and allocated shares. 3. The expense is available for payment tracking and settlement.
Normal Flow	1. The user selects the Create Shared Expense option. 2. The system displays the shared expense form. 3. The user enters the title, total amount, date, optional description, and, optionally, the participant who paid the bill. 4. The user submits the form. 5. The system validates the entered information. 6. The system saves the shared expense and records the user as its creator. 7. The system includes the Add Participants use case (UC-10). 8. The system includes the Split Shared Bill use case (UC-11). 9. The system displays a confirmation message and the shared expense details.
Alternative Flows	A1. Creation Cancelled: 1. The user cancels before submitting. 2. The system discards the entered data and no expense is created.
Exceptions	E1. Missing Required Information: The system identifies the missing fields and asks the user to complete them. E2. Invalid Amount or Date: The system displays a validation message and asks the user to correct the value (e.g., amount must be greater than zero). E3. No Valid Participants: If fewer than two participants (including the creator) are provided, the system requests valid participants and does not complete the creation. E4. Saving Failure: The system displays an error message and does not indicate that the shared expense was created.
Includes / Extends	«include» Add Participants (UC-10); «include» Split Shared Bill (UC-11)
Priority	High
Frequency of Use	A few times per month per user.
Business Rules	The creator is added as a participant automatically. Expense, participants, and shares are saved in one transaction (NFR-04). Only the creator may change the title, total, or participants, until the first settlement is confirmed (FR-09).

UC-10: Add Participants

Use Case ID	UC-10
Use Case Name	Add Participants
Related FR	FR-10
Primary Actor	Registered User (creator)
Secondary Actors	None

Description	This use case describes how the creator of a shared expense selects the registered users who take part in that expense. It is included by the Create Shared Expense use case.
Trigger	The system reaches the participant selection step during Create Shared Expense, or the creator opens an existing shared expense and chooses Add Participants.
Preconditions	1. The user is signed in to FinPact. 2. The shared expense exists. 3. The user is the creator of the shared expense (or otherwise authorized to manage it).
Postconditions	1. The selected users are associated with the shared expense as participants. 2. The updated participant list is displayed.
Normal Flow	1. The user opens the participant selection step of the shared expense. 2. The system displays an email field and the current participant list (the creator is already included). 3. The user enters the email address of a participant. 4. The system finds the registered user by exact email match and shows only that user's name. 5. The user confirms; steps 3–5 repeat for each participant. 6. The system validates the list (no duplicates, all registered users, 2–20 participants including the creator). 7. The system associates the participants with the shared expense and displays the updated list.
Alternative Flows	A1. Remove Participant Before Split: 1. The user removes a previously selected participant. 2. The system updates the list and re-validates the minimum participant count.
Exceptions	E1. Duplicate Participant: The system prevents the same user from being added twice to the same expense and informs the user. E2. Ineligible or Unknown User: If the selected person is not a registered FinPact user, the system explains the problem and does not add them. E3. Unauthorized Operation: If the user is not authorized to manage the expense, the system denies the operation. E4. Saving Failure: The system displays an error message and keeps the previous participant list.
Includes / Extends	Included by UC-09
Priority	High
Frequency of Use	With every shared expense.
Business Rules	2–20 participants including the creator; participants are found by exact email match and the full user list is never shown (FR-10, C-12).

UC-11: Split Shared Bill

Use Case ID	UC-11

Use Case Name	Split Shared Bill
Related FR	FR-11
Primary Actor	Registered User (creator)
Secondary Actors	None
Description	This use case describes how the system calculates each participant's share of a shared expense using an equal split or a custom split whose amounts add up to the total. It is included by the Create Shared Expense use case.
Trigger	The system reaches the split step during Create Shared Expense, or the creator selects the Split option on an existing shared expense.
Preconditions	1. The user is signed in to FinPact. 2. The shared expense exists and has a valid total amount. 3. The shared expense has at least two participants.
Postconditions	1. A share amount is saved for every participant. 2. The sum of all shares equals the total amount of the bill. 3. The shares are available for balance tracking.
Normal Flow	1. The user opens the split step of the shared expense. 2. The system displays the total amount and the participants. 3. The user selects the Equal Split method. 4. The system divides the total amount by the number of participants (e.g., EGP 1,200 4 = EGP 300). 5. The system rounds each share down to two decimals and adds the remaining piasters to the creator's share (e.g., EGP 100.00 for 3 → 33.33, 33.33, 33.34). 6. The system displays the total and each participant's share. 7. The user confirms the split. 8. The system saves the share amount of each participant.
Alternative Flows	A1. Custom Split: 1. The user selects the Custom Split method. 2. The user enters an amount for each participant. 3. The system checks that the amounts add up to the total. 4. The system saves the custom shares. A2. Recalculate Split: 1. The participant list changes before settlement. 2. The system recalculates the shares and asks the user to confirm.
Exceptions	E1. No Participants: The system cannot calculate shares and asks the user to add valid participants. E2. Custom Amounts Do Not Match Total: The system shows the difference, rejects the split, and asks the user to correct the amounts. E3. Invalid Bill Amount: If the total is zero, negative, or invalid, the system asks the user to correct it. E4. Saving Failure: The system displays an error message and does not indicate that the split was saved.

Includes / Extends	Included by UC-09
Priority	High
Frequency of Use	With every shared expense.
Business Rules	Equal split rounds each share down to two decimals and adds the remaining piasters to the creator's share (A-14); custom shares must add up exactly to the total (FR-11).

UC-12: Track Payments and Amounts Owed

Use Case ID	UC-12
Use Case Name	Track Payments and Amounts Owed
Related FR	FR-12
Primary Actor	Registered User (creator or participant)
Secondary Actors	None
Description	This use case describes how a user records who paid a shared expense and views the remaining amount each participant owes, based on the bill total, each participant's share, and the recorded payments.
Trigger	The user opens a shared expense and selects the Payments and Balances option.
Preconditions	1. The user is signed in to FinPact. 2. The shared expense exists, has participants, and has saved shares. 3. The user is the creator or a participant of the shared expense.
Postconditions	1. The payment is recorded as a payment record linked to the shared expense and the payer. 2. The outstanding amount of each participant is calculated and displayed.
Normal Flow	1. The user opens the shared expense details. 2. The user selects the Payments and Balances option. 3. The system displays the total amount, participant shares, and any existing payments. 4. The user records a payment by selecting the payer and entering the amount paid and the payment date. 5. The system validates the payment information. 6. The system saves the payment record. 7. The system calculates the balance of each participant from the bill total, their share, and the recorded payments. 8. The system displays who paid and how much each participant still owes (e.g., if one person pays EGP 1,200 and the shares are EGP 300 each, each of the other three owes EGP 300 to the payer).

Alternative Flows	A1. View Balances Only: 1. The user opens the balances view without adding a payment. 2. The system displays the current payments and outstanding amounts. A2. Multiple Payers: 1. The user records payments made by more than one participant. 2. The system recalculates each participant's balance accordingly.
Exceptions	E1. Missing Payment Information: The system asks the user to complete the missing payer, amount, or date. E2. Invalid Payment Amount: If the amount is zero, negative, or exceeds the bill total, the system displays a validation message and does not save it. E3. Payments Exceed Total: If the new payment would make the total of payments exceed the bill total, the system rejects it and shows the remaining unpaid amount. E4. Unauthorized Access: If the user is not the creator or a participant, the system denies access. E5. Saving Failure: The system displays an error message and does not indicate that the payment was recorded.
Includes / Extends	None
Priority	High
Frequency of Use	Weekly for active groups.
Business Rules	The payer must be a participant; total payments cannot exceed the bill total; net balance = paid − share + confirmed settlements sent − confirmed settlements received (FR-12).

UC-13: View Shared Expense History

Use Case ID	UC-13
Use Case Name	View Shared Expense History
Related FR	FR-13
Primary Actor	Registered User (creator or participant)
Secondary Actors	None
Description	This use case describes how a user views the shared expenses they created or participate in, and opens the details of one of them.
Trigger	The user selects Shared Expenses from the navigation bar.
Preconditions	1. The user is signed in to FinPact.
Postconditions	1. The user's shared expenses (and, if selected, one expense's details) are displayed. 2. No stored data is modified.

Normal Flow	1. The user selects Shared Expenses. 2. The system retrieves the shared expenses where the user is the creator or a participant. 3. The system displays each expense's title, date, total, the user's share, the user's net balance, and status, newest first. 4. The user selects an expense. 5. The system displays its participants and shares, payments, settlements with status, and the "who owes whom" list.
Alternative Flows	A1. Filter by Status: 1. The user selects Open or Settled. 2. The system displays only the matching expenses.
Exceptions	E1. No Shared Expenses: The system displays "No shared expenses yet" and a link to Create Shared Expense. E2. Unauthorized Access: If the user requests an expense they are not part of, the system denies access. E3. Retrieval Failure: The system displays an error message instead of partial or misleading data.
Includes / Extends	None
Priority	High
Frequency of Use	Weekly.
Business Rules	Only shared expenses in which the user is creator or participant are listed (A-13).

UC-14: Record Settlement

Use Case ID	UC-14
Use Case Name	Record Settlement
Related FR	FR-20
Primary Actor	Registered User (debtor)
Secondary Actors	Receiving Participant (confirms or rejects)
Description	This use case describes how a participant who owes money records that they have settled an amount with another participant, and how the receiver confirms it. No real money is transferred.
Trigger	The debtor selects Record Settlement on a shared expense in which they owe money.
Preconditions	1. The debtor is signed in and is a participant with a negative balance. 2. The receiver is a participant of the same expense with a positive balance.
Postconditions	1. A settlement is stored with status Pending. 2. After confirmation, its status is Confirmed and both balances are updated. 3. If all balances are zero, the expense status becomes Settled.

Normal Flow	1. The debtor opens the shared expense details. 2. The debtor selects Record Settlement. 3. The system displays the form, pre-filled with the receiver and amount from the “who owes whom” list, and a notice that FinPact does not transfer money. 4. The debtor confirms or adjusts the receiver and amount and submits. 5. The system validates that the amount is positive and does not exceed the debtor’s debt or the receiver’s credit. 6. The system saves the settlement as Pending with the current server date. 7. The receiver opens the expense and selects Confirm on the pending settlement. 8. The system sets the status to Confirmed and recalculates the balances. 9. If every balance is zero, the system marks the expense as Settled. 10. The system displays a confirmation to both parties.
Alternative Flows	A1. Partial Settlement: The debtor settles less than the full debt; the remaining amount stays outstanding. A2. Receiver Rejects: 1. The receiver selects Reject. 2. The system sets the status to Rejected; balances do not change, and the debtor sees the rejection.
Exceptions	E1. Invalid Amount: If the amount is zero, negative, or larger than the debt or credit, the system shows a validation message and saves nothing. E2. No Outstanding Debt: The Record Settlement option is not available to a participant whose balance is zero or positive. E3. Unauthorized User: A user who is not a participant cannot record, confirm, or reject a settlement. E4. Saving Failure: The system shows an error message and the balances remain unchanged.
Includes / Extends	None
Priority	High
Frequency of Use	Once or twice per shared expense.
Business Rules	Only Confirmed settlements change balances; the amount cannot exceed the debtor’s debt or the receiver’s credit; no money is transferred (FR-20, C-05).

UC-15: Set Monthly Budget

Use Case ID	UC-15
Use Case Name	Set Monthly Budget
Related FR	FR-14
Primary Actor	Registered User
Secondary Actors	None

Description	This use case describes how a user sets or updates the total spending budget for a calendar month.
Trigger	The user selects Set Monthly Budget on the Budgets page.
Preconditions	1. The user is signed in to FinPact.
Postconditions	1. A budget exists for the selected month with the entered total. 2. Budget usage is recalculated.
Normal Flow	1. The user opens the Budgets page and selects Set Monthly Budget. 2. The system displays the form with the month (default: current) and amount fields. 3. The user enters the amount and submits. 4. The system validates the amount and checks that the month is current or future. 5. The system checks that no budget exists for that month and creates one. 6. The system recalculates usage (FR-16) and checks thresholds (FR-17). 7. The system displays a confirmation and the budget progress bar.
Alternative Flows	A1. Update Existing Budget: 1. At step 5 the system finds an existing budget for the month. 2. The system shows the current value and asks the user to confirm the update. 3. The system updates the budget and continues at step 6.
Exceptions	E1. Invalid Amount: The system asks the user to enter a valid positive amount. E2. Past Month: The system explains that past budgets are read-only and saves nothing. E3. Below Category Limits: If the total is lower than the sum of the month's category limits, the system rejects it and shows that sum. E4. Saving Failure: The system shows an error message and keeps the previous budget.
Includes / Extends	None
Priority	High
Frequency of Use	Monthly.
Business Rules	One budget per user and month; past months are read-only; the total cannot be lower than the sum of its category limits (FR-14).

UC-16: Set Category Spending Limits

Use Case ID	UC-16
Use Case Name	Set Category Spending Limits
Related FR	FR-15
Primary Actor	Registered User
Secondary Actors	None

Description	This use case describes how a user assigns a spending limit to an expense category within a monthly budget.
Trigger	The user selects Set Category Limit on a monthly budget.
Preconditions	1. The user is signed in to FinPact. 2. A budget exists for the selected month.
Postconditions	1. The category limit is stored for the budget. 2. Category usage is recalculated.
Normal Flow	1. The user opens the budget for a month and selects Set Category Limit. 2. The system displays the category list and an amount field. 3. The user selects a category, enters the limit, and submits. 4. The system validates the amount. 5. The system checks that the sum of all category limits does not exceed the monthly total. 6. The system saves the limit, replacing any existing limit for that category. 7. The system recalculates category usage and checks thresholds (FR-17). 8. The system displays the category progress bar and a confirmation.
Alternative Flows	A1. Remove Limit: 1. The user selects Remove on a category limit and confirms. 2. The system deletes the limit; the category then counts only towards the monthly total.
Exceptions	E1. No Monthly Budget: The system asks the user to set a monthly budget first (UC-15). E2. Invalid Amount: The system asks for a valid positive amount. E3. Limits Exceed Budget: The system rejects the limit and shows the remaining amount available. E4. Saving Failure: The system shows an error message and keeps the previous limit.
Includes / Extends	None
Priority	High
Frequency of Use	Monthly.
Business Rules	One limit per category and budget; the sum of limits cannot exceed the monthly total (FR-15).

UC-17: Monitor Budget Usage

Use Case ID	UC-17
Use Case Name	Monitor Budget Usage
Related FR	FR-16
Primary Actor	Registered User
Secondary Actors	None

Description	This use case describes how a user reviews how much of the monthly budget and each category limit has been spent. It is extended by Receive Budget Warning.
Trigger	The user opens the Budgets page or the Dashboard.
Preconditions	1. The user is signed in to FinPact.
Postconditions	1. Spent, remaining, and usage percentage are displayed. 2. No stored data is modified.
Normal Flow	1. The user opens the Budgets page. 2. The system selects the current month. 3. The system retrieves the month's budget, category limits, and personal expenses. 4. The system calculates spent, remaining, and usage % for the total and for each limited category. 5. The system displays progress bars in the Normal, Warning, or Exceeded state. Extension point: when usage reaches 80% or 100%, Receive Budget Warning (UC-18) applies.
Alternative Flows	A1. Another Month: The user selects a different month; the system repeats steps 3–5 for it. A2. No Category Limits: Only the monthly total is displayed.
Exceptions	E1. No Budget: The system displays "No budget set for this month" and a link to Set Monthly Budget. E2. Retrieval Failure: The system shows an error message instead of incorrect figures.
Includes / Extends	Extended by UC-18
Priority	High
Frequency of Use	Several times per week.
Business Rules	Usage is calculated from personal expenses only (A-15); usage % = spent ÷ amount × 100, one decimal place (FR-16).

UC-18: Receive Budget Warning

Use Case ID	UC-18
Use Case Name	Receive Budget Warning
Related FR	FR-17
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how the system warns the user when spending reaches 80% or 100% of the monthly budget or a category limit. It extends Monitor Budget Usage under the condition [spending reaches a threshold].

Trigger	Saving an expense, or lowering a budget or limit, makes usage reach a threshold.
Preconditions	1. The user is signed in. 2. A budget (and, for a category warning, a category limit) exists for the month. 3. No notification for that threshold exists yet for that budget or limit.
Postconditions	1. An unread budget notification is stored. 2. The warning is displayed to the user.
Normal Flow	1. The user saves an expense (UC-04, UC-05) or changes a budget or limit (UC-15, UC-16). 2. The system recalculates usage. 3. The system detects that usage reached 80% or 100% of the total or of a category limit. 4. The system confirms that no notification exists for that threshold. 5. The system stores a notification with message, threshold, creation time, and isRead = false. 6. The system displays a warning banner. 7. The system adds the notification to the list and updates the unread count on the Dashboard.
Alternative Flows	A1. Mark as Read: 1. The user opens the notification list and selects Mark as Read. 2. The system sets isRead = true and decreases the unread count. A2. Both Thresholds Passed at Once: Only the 100% notification is created.
Exceptions	E1. Notification Saving Failure: The expense or budget change remains saved; the banner is still shown from the calculated usage, and the notification is created at the next recalculation.
Includes / Extends	«extend» Monitor Budget Usage (UC-17) [usage ≥ 80% or ≥ 100%]
Priority	High
Frequency of Use	A few times per month.
Business Rules	At most one notification per threshold, budget or limit, and month; if both thresholds are passed at once, only the 100% notification is stored (FR-17).

UC-19: View Spending Analytics

Use Case ID	UC-19
Use Case Name	View Spending Analytics
Related FR	FR-18
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a user views a breakdown of spending by category for a selected period.
Trigger	The user selects Analytics from the navigation bar.

Preconditions	1. The user is signed in to FinPact.
Postconditions	1. The spending summary is displayed. 2. No stored data is modified.
Normal Flow	1. The user opens the Analytics page. 2. The system selects the current month as the period. 3. The system retrieves the user's personal expenses in the period. 4. The system calculates the period total and each category's total, percentage, and number of expenses. 5. The system displays a chart and a table sorted by amount.
Alternative Flows	A1. Change Period: 1. The user selects another month or a custom date range. 2. The system repeats steps 3–5 for that period.
Exceptions	E1. No Expenses: The system displays "No expenses in this period". E2. Invalid Range: If the start date is after the end date or the range exceeds 12 months, the system asks the user to correct it. E3. Retrieval Failure: The system shows an error message.
Includes / Extends	None
Priority	Medium
Frequency of Use	Weekly to monthly.
Business Rules	Only personal expenses are analysed; a custom range cannot exceed 12 months (FR-18).

UC-20: View Spending Summary and Trends

Use Case ID	UC-20
Use Case Name	View Spending Summary and Trends
Related FR	FR-19
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a user compares spending between two months and views the six-month spending trend.
Trigger	The user selects the Trends tab on the Analytics page.
Preconditions	1. The user is signed in to FinPact.
Postconditions	1. The comparison and trend chart are displayed. 2. No stored data is modified.

Normal Flow	1. The user selects the Trends tab. 2. The system selects the current and previous months. 3. The system calculates, for each category and in total, the amount in each month, the difference, and the change percentage. 4. The system calculates the monthly totals for the last six months. 5. The system displays the comparison table and the trend chart.
Alternative Flows	A1. Choose Months: 1. The user selects two other months. 2. The system repeats step 3 and updates the table.
Exceptions	E1. No Data: If neither month has expenses, the system displays a message instead of an empty chart. E2. Same Month Twice: The system asks the user to choose two different months. E3. Retrieval Failure: The system shows an error message.
Includes / Extends	None
Priority	Medium
Frequency of Use	Monthly.
Business Rules	Change % = (B − A) ÷ A × 100; shown as “New” when the earlier month is zero (FR-19).

UC-21: View Expense History

Use Case ID	UC-21
Use Case Name	View Expense History
Related FR	FR-06
Primary Actor	Registered User
Secondary Actors	None
Description	This use case describes how a user views their recorded personal expenses. It belongs to the Account and Personal Expense Management diagram and is extended by Search and Filter Expenses (UC-08).
Trigger	The user selects Personal Expenses from the navigation bar.
Preconditions	1. The user is signed in to FinPact.
Postconditions	1. The user’s expenses are displayed. 2. No stored data is modified.
Normal Flow	1. The user selects Personal Expenses. 2. The system retrieves the user’s own expenses. 3. The system displays amount, date, category, and description, newest first, 20 per page, with the total of the listed expenses. 4. The user may select an expense to edit (UC-05) or delete (UC-06). Extension point: Search and Filter Expenses (UC-08).

Alternative Flows	A1. Next Page: The user moves to another page; the system displays the next 20 expenses.
Exceptions	E1. No Expenses: The system displays “No expenses recorded yet” and a link to Add Expense. E2. Retrieval Failure: The system shows an error message instead of an empty list.
Includes / Extends	Extended by UC-08
Priority	High
Frequency of Use	Daily.
Business Rules	Only the user’s own expenses, newest first, 20 per page (FR-06, CR-2).

3.3 Conceptual Class Model

The conceptual class model shows the domain entities that the Model layer of the MVC architecture (C-02) implements, their key attributes, and the associations between them. Foreign keys, data types, and value constraints are given once in Appendix A and are not repeated here.

3.3.1 Class Diagram

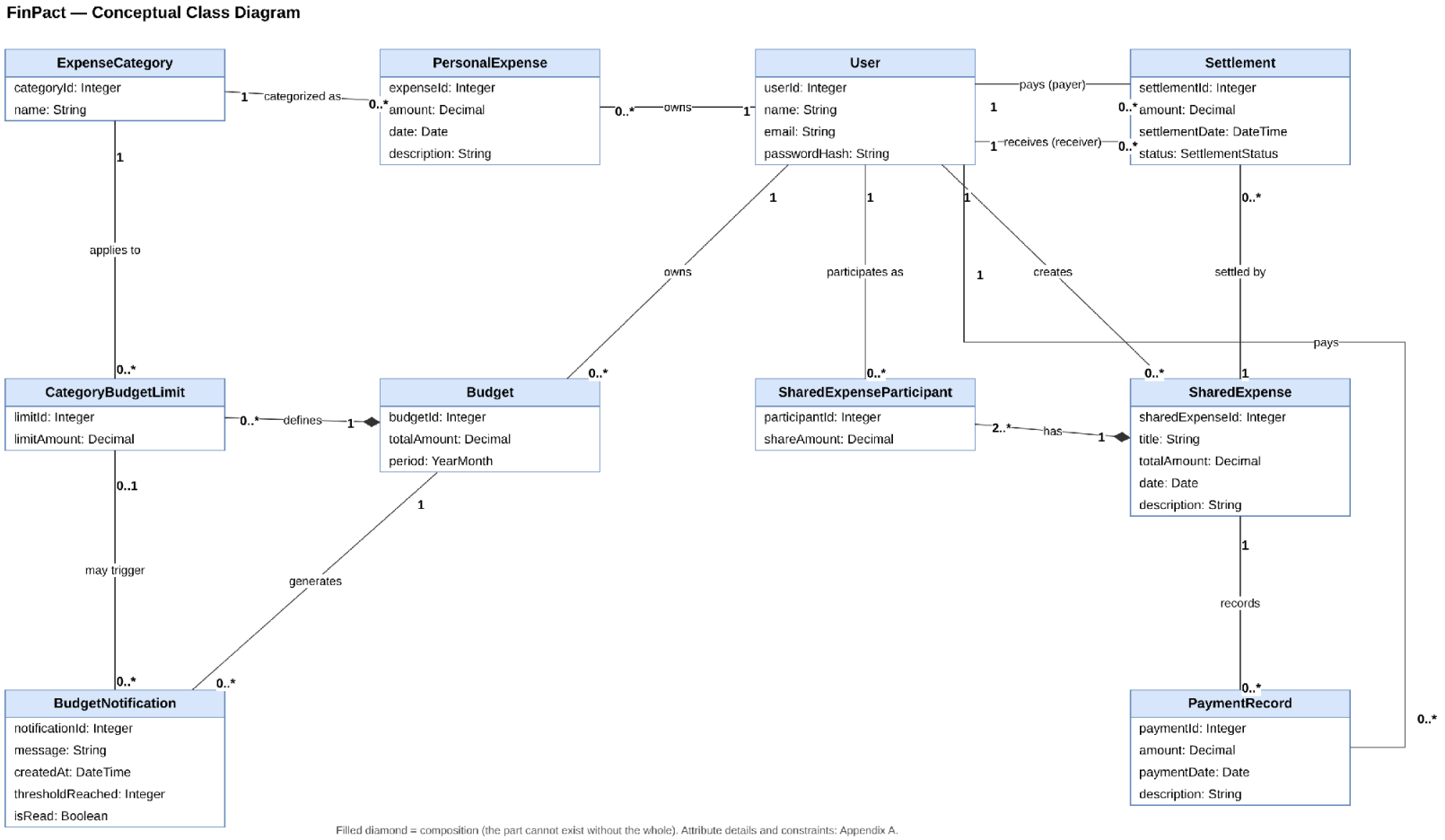


Figure 5: FinPact Conceptual Class Diagram

3.3.2 Class Descriptions

Class	Responsibility	Associations (multiplicity)
User	A registered account holder. Authenticates and owns all personal data.	Owns 0..* PersonalExpense and 0..* Budget; creates 0..* SharedExpense; participates as 0..* SharedExpenseParticipant; pays 0..* PaymentRecord; is payer or receiver of 0..* Settlement.
PersonalExpense	One spending record of a single user (amount, date, description).	Belongs to exactly 1 User; categorized as exactly 1 ExpenseCategory.
ExpenseCategory	One entry of the fixed category list (A-16).	Categorizes 0..* PersonalExpense; is limited by 0..* CategoryBudgetLimit.
SharedExpense	A bill shared by 2–20 participants; its status (Open / Settled) is derived from balances.	Created by 1 User; composed of 2..* SharedExpenseParticipant; records 0..* PaymentRecord; settled by 0..* Settlement.
SharedExpenseParticipant	The link between a user and a shared expense, holding that user's share.	Part of exactly 1 SharedExpense (composition); refers to exactly 1 User.
PaymentRecord	An amount actually paid towards a shared expense.	Belongs to 1 SharedExpense; paid by 1 User, who must be a participant.
Settlement	A debtor's record of settling an amount with a receiver; status Pending, Confirmed, or Rejected.	Belongs to 1 SharedExpense; has exactly 1 payer User and 1 receiver User.
Budget	The total spending limit of one user for one calendar month.	Owned by 1 User; composed of 0..* CategoryBudgetLimit; generates 0..* BudgetNotification.
CategoryBudgetLimit	A limit for one category inside a monthly budget.	Part of exactly 1 Budget (composition); applies to 1 ExpenseCategory; may trigger 0..* BudgetNotification.
BudgetNotification	A stored 80% or 100% warning, read or unread.	Generated for 1 Budget; refers to 0..1 CategoryBudgetLimit (none for a warning on the monthly total).

Derived values: net balance, shared-expense status, spent, remaining, and usage % are calculated by the Models and are not stored (Appendix A.11).

3.3.3 State Diagrams

Three objects change state in ways that affect what users may do next. Figure 6 (next page) shows their life cycles; each transition is triggered by a use case in Section 3.2.

Settlement (FR-20, UC-14): created as Pending by the debtor. The receiver either confirms it (Confirmed, balances updated) or rejects it (Rejected, balances unchanged). Both end states are kept as history and cannot be changed.

Shared Expense (FR-09, FR-12, UC-09, UC-12, UC-14): Open from creation. Payments and confirmed settlements keep it Open while any balance is non-zero; it becomes Settled when payments equal the total and every net balance is zero.

Budget or Category Limit Usage (FR-16, FR-17, UC-17, UC-18): Normal below 80%, Warning from 80% to 99.9%, Exceeded at 100% or more. Entering Warning or Exceeded stores one notification; a jump from Normal straight to Exceeded stores only the 100% notification. Editing or deleting expenses can move usage back to a lower state without deleting earlier notifications.

FinPact — State Diagrams of Key Objects

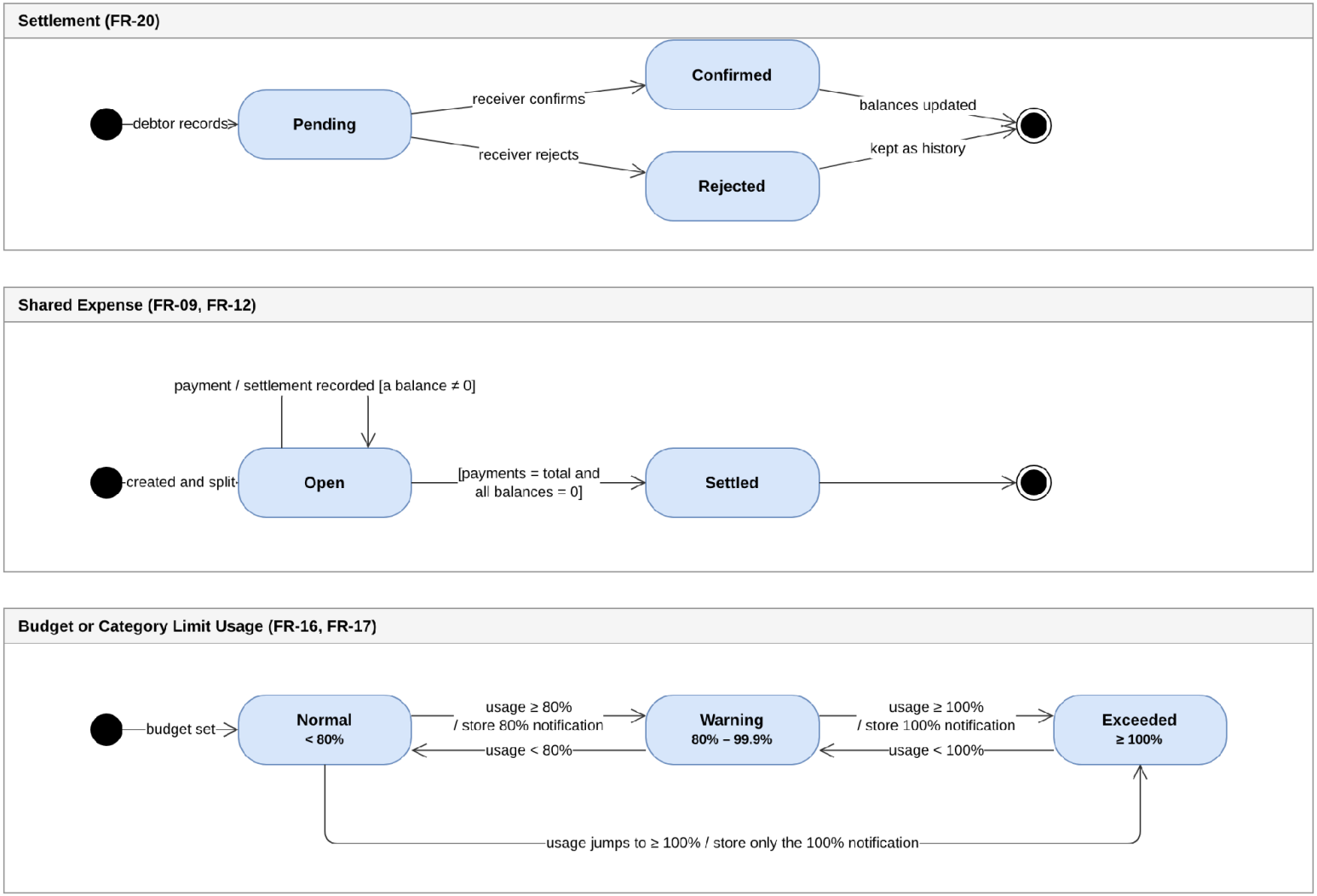


Figure 6: State Diagrams of Settlement, Shared Expense, and Budget Usage (made via Draw.io)

3.4 Non-Functional Requirements

Each non-functional requirement states a measurable target, the method used to verify it, and its acceptance criteria. Test evidence is stored in the repository's /tests folder.

3.4.1 NFR-01: Performance

Requirement

Under a workload of 10 concurrent users and the data scale of Section 2.5.6, at least 95% of backend requests for each of the following core operations shall complete within 3 seconds:

- Viewing personal expense history (FR-06).
- Saving a new personal expense (FR-03).
- Viewing the Dashboard (FR-12, FR-16, FR-17 summaries).
- Viewing shared expense details (FR-13).

The target applies to backend response time (request sent → response received), not to full page rendering in the browser.

Verification Method: Apache JMeter

- Seed the test database with representative data (about 100 users, 1,000 personal expenses for the test user, 100 shared expenses).
- Configure a JMeter test plan with the four endpoints, their HTTP methods, parameters, and a logged-in session for each simulated user.
- Run a smoke test to confirm each request returns the expected response.
- Run the plan with 10 concurrent threads, at least 100 requests per operation.
- Record average and 95th-percentile response times per operation; investigate any failed requests rather than counting them as fast responses.
- Record the environment (hardware, application version, database state) so the test can be repeated.

Acceptance Criteria

- The 95th-percentile response time of each of the four operations is ≤ 3 seconds.
- The error rate during the run is 0%, or every failure is explained and fixed before re-testing.
- The JMeter plan and results are committed to /tests with a pass/fail verdict per operation.

3.4.2 NFR-02: Security and Access Control

Requirement

FinPact shall protect accounts and financial data against unauthorized access and modification:

- Every protected page and endpoint requires authentication (CR-1, C-07).

- Authorization is checked on the server for every request, so that users can access only the records allowed by CR-2 and C-12.

- Passwords are stored only as salted hashes (C-04); sessions are invalidated on logout and after 30 minutes of inactivity (FR-02).

- Database access uses parameterized queries or an ORM, preventing SQL injection.
- Error messages reveal no passwords, other users' data, or internal details (C-14).

Verification Method

- **Authentication test:** request each protected page and endpoint without logging in; expect a redirect to login or an unauthorized response.
- **Authorization test:** with two test accounts A and B, have B request, edit, and delete A's personal expenses, budgets, and a shared expense B is not part of; expect rejection and unchanged records.
- **Password storage inspection:** inspect the users table and confirm that only hashes are stored.
- **Session test:** log in, log out, then replay the old session cookie; expect access to be denied. Repeat after 30 minutes of inactivity.
- **Injection and error test:** submit SQL-injection strings (e.g., ' OR '1'='1) in login, search, and form fields, plus invalid credentials; expect normal validation errors and no internal details.

Acceptance Criteria

- 100% of unauthenticated and cross-user access attempts are rejected, and the targeted records remain unchanged.
- No password is stored in plaintext.
- A logged-out or expired session cannot access any protected resource.
- No injection attempt alters data or returns database errors, and no error response discloses internal details.

3.4.3 NFR-03: Usability

Requirement

FinPact shall provide a consistent, understandable interface: clearly labelled form fields, consistent navigation and buttons, validation messages that name the field and the problem, feedback after every action, confirmation before destructive actions, and financial figures (spent, remaining, owed, owed to you) that are clearly labelled. Basic accessibility practices follow C-11.

Verification Method

At least five test users who are familiar with web applications but have not used FinPact perform the following tasks without help, while the tester records completion, time, errors, and points of confusion:

- Register and log in.
- Add a personal expense.

-Find a recorded expense using search or filters.

-Create a shared expense with three participants and an equal split.

-Set a monthly budget and a category limit, then state the amount spent and remaining.

- Read a validation message and correct the invalid input.

- Delete an expense (the tester notes whether a confirmation dialog appears).

-Identify how much they owe in a shared expense and record a settlement.

Acceptance Criteria

- At least 90% of participants complete each task without assistance.

-Adding a personal expense takes on average 60 seconds or less; creating a shared expense takes on average 2 minutes or less.

-Participants correctly read the budget figures and their balance, and correct invalid input using the displayed messages.

-Every destructive action tested asks for confirmation.

-Any issue that blocks a task is fixed and re-tested before final delivery.

3.4.4 NFR-04: Reliability and Data Integrity

Requirement

FinPact shall keep stored records and calculated values consistent and correct:

-Shares always add up to the shared expense total, and payments never exceed it (FR-11, FR-12).

-Balances, budget usage, and remaining amounts are always derived from the current records and reflect every successful change.

-Operations that change several records (e.g., creating a shared expense with participants and shares, confirming a settlement) run in a single database transaction; if any step fails, no change is saved.

-A failed operation never displays a success message (CR-6).

Verification Method

Automated unit and integration tests in /tests, each with expected results defined before execution:

Calculation tests: known inputs for equal and custom splits (including rounding, e.g., EGP 100 / 3), balances, budget usage, and trend percentages.

Validation tests: invalid amounts, missing fields, custom shares that do not match the total, payments above the total, and settlements above the debt; expect rejection and unchanged records.

Recalculation tests: add, edit, and delete expenses and confirm settlements, then verify the updated usage and balances.

Failure tests: force a database error in the middle of a multi-record operation and verify that no partial data remains and no success message is shown.

Acceptance Criteria

100% of calculation, validation, recalculation, and failure test cases pass.

No test leaves the database in an inconsistent state (e.g., a shared expense without shares).

3.4.5 NFR-05: Browser Compatibility and Responsive Layout

Requirement

All core functions shall work, with no overlapping, clipped, or unreachable elements and no unintended horizontal scrolling, in the following configurations:

Browsers: Google Chrome, Mozilla Firefox, and Microsoft Edge (latest stable versions at test time).

Viewports: 360 × 640 (phone), 768 × 1024 (tablet), 1366 × 768 (laptop), and 1920 × 1080 (desktop).

On narrow viewports the navigation bar collapses into a menu and wide tables scroll inside their own container.

Verification Method

- Run a checklist of core test cases, log in, add/edit/delete an expense, filter expenses, create a shared expense, record a settlement, set a budget, and view analytics in every browser/viewport combination (3 × 4 = 12).

-Use browser developer tools' device mode for the viewports, plus at least one real Android or iOS phone.

-Record browser versions, viewports, results, and screenshots of any defects.

Acceptance Criteria

-All core test cases pass in all 12 configurations.

-No layout defect prevents completing a core task; all detected defects are documented and fixed.

3.5 External Interface Requirements

3.5.1 User Interfaces

FinPact provides a browser-based, responsive, English-language UI (C-08, C-09) that uses consistent navigation, labeled form fields, and clear validation messages. The main UI elements are:

Registration page: a form with name, email, password, and password-confirmation fields and a Register button. Validation messages appear next to invalid fields, and a link leads to the login page.

Login page: email and password fields, a Log In button, and a link to registration. Failed logins show a generic error message that does not reveal which field was wrong.

Navigation bar: shown only to signed-in users. It links to the Dashboard, Personal Expenses, Shared Expenses, Budgets, Analytics, and Notifications (with an unread count), and includes a Log Out button. On mobile viewports it collapses into a menu.

Dashboard: a summary of the current month's spending, budget usage, outstanding balances (amounts the user owes or is owed), and unread budget notifications.

Personal Expenses page: a table or list of expenses (amount, date, category, description) with a search box, category/date/amount filters, and Add, Edit, and Delete actions. Deletion requires a confirmation dialog.

Expense form: fields for amount, date, category (drop-down list), and description, with Save and Cancel buttons.

Shared Expenses page: a list of shared bills the user created or participates in, showing title, total amount, date, and the user's balance status. A bill details view shows participants, each share, payments made, and settlements recorded.

Create Shared Expense form: fields for title, total amount, date, and description. It includes a participant selector and a choice between equal and custom split. For a custom split, the entered shares are checked against the total before saving.

Settlement form: lets a debtor choose the receiver and amount (pre-filled from the "who owes whom" list) and submit a pending settlement; the receiver sees Confirm and Reject buttons. A notice states that FinPact does not transfer real money.

Budgets page: forms to set the monthly budget and per-category limits. Progress bars show spending, remaining amount, and usage percentage per budget and per category.

Budget notifications: warning banners or a notification list shown when usage reaches a threshold (e.g., 80% or 100%). Each can be marked as read.

Analytics page: category and period summaries and spending-trend comparisons, displayed as tables and/or charts with a period selector.

Help and feedback: contextual tooltips, a link to the User Guide and FAQ, and success/error messages after each operation. Error messages never expose internal details (C-14).

3.5.2 Hardware Interfaces

FinPact has no direct hardware interfaces. It does not interact with printers, scanners, sensors, or other peripherals. Users access it through the standard input and display devices of their laptop, desktop, tablet, or smartphone (keyboard, mouse, or touchscreen). The server-side hardware is managed by the hosting environment.

3.5.3 Software Interfaces

Database: the Model layer communicates with a relational database management system (e.g., MySQL or PostgreSQL) to store users, personal expenses, categories, shared expenses, participants, payment records, budgets, category limits, settlements, and notifications. Access goes through the application's data-access layer, using parameterized queries or an ORM to prevent SQL injection.

Web framework / runtime: FinPact runs on the course-approved server-side runtime and MVC framework (C-01, C-02). Controllers receive requests from the browser and call the Models, and Views render the responses.

Authentication and security libraries: a maintained password-hashing library (e.g., bcrypt or Argon2) and a session-management mechanism handle secure login and logout (C-04, C-07).

Client-side libraries: standard browser technologies (HTML, CSS, JavaScript) and, optionally, a charting library for analytics (D-05).

External systems: FinPact does not integrate with payment gateways, banking APIs, email services, or third-party APIs, and it exposes no public API in Phase 2 (C-05).

3.5.4 Communication Interfaces

Protocol: all communication between the user's browser and the FinPact server uses HTTPS (HTTP over TLS), so data in transit is encrypted.

Request model: the application uses standard HTTP request/response semantics (GET for retrieving pages and data, POST/PUT/DELETE for changing data). Protected requests carry a session cookie or equivalent token.

Server-database communication: the server connects to the database over the DBMS's internal protocol, over a private or internal connection that is not exposed to end users.

Data format: pages are delivered as HTML, with CSS and JavaScript resources and, where needed, JSON for dynamic updates.

Other protocols: FTP, SMTP, and proprietary protocols are not used in Phase 2.

4. Appendices

Appendix A: Data Dictionary

This appendix defines every attribute of the conceptual class model (Section 3.3) with its data type and constraints. Foreign-key attributes (marked FK) are not drawn in the class diagram; they implement its associations and are listed here so the database design can be derived directly. PK = primary key. “Decimal(12,2)” means two decimal places (CR-3).

A.1 User

Attribute	Type	Constraints / Description
userId	Integer	PK, auto-generated.
name	String(50)	Required; 2–50 characters.
email	String(100)	Required; valid format; unique (case-insensitive); stored in lowercase.
passwordHash	String(255)	Required; salted hash (bcrypt/Argon2) of a password of ≥ 8 characters with a letter and a digit (C-04).

A.2 PersonalExpense

Attribute	Type	Constraints / Description
expenseId	Integer	PK, auto-generated.
amount	Decimal(12,2)	Required; 0.01–1,000,000.00.
date	Date	Required; not later than today (server date).
description	String(255)	Optional.
userId (FK)	Integer	Required; owner → User.userId.
categoryId (FK)	Integer	Required; → ExpenseCategory.categoryId.

A.3 ExpenseCategory

Attribute	Type	Constraints / Description
categoryId	Integer	PK.
name	String(30)	Required; unique; one of Food, Transportation, Shopping, Bills, Entertainment, Education, Other (A-16).

A.4 SharedExpense

Attribute	Type	Constraints / Description
sharedExpenseId	Integer	PK, auto-generated.
title	String(100)	Required; 1–100 characters.

Attribute	Type	Constraints / Description
totalAmount	Decimal(12,2)	Required; 0.01–1,000,000.00; equals the sum of all participants' shareAmount.
date	Date	Required; not later than today.
description	String(255)	Optional.
creatorId (FK)	Integer	Required; → User.userId; the creator is also a participant.

A.5 SharedExpenseParticipant

Attribute	Type	Constraints / Description
participantId	Integer	PK, auto-generated.
shareAmount	Decimal(12,2)	Required; ≥ 0.01.
sharedExpenseId (FK)	Integer	Required; → SharedExpense; 2–20 participants per expense.
userId (FK)	Integer	Required; → User; (sharedExpenseId, userId) is unique.

A.6 PaymentRecord

Attribute	Type	Constraints / Description
paymentId	Integer	PK, auto-generated.
amount	Decimal(12,2)	Required; > 0; sum of payments per expense ≤ totalAmount.
paymentDate	Date	Required; not later than today.
description	String(255)	Optional.
sharedExpenseId (FK)	Integer	Required; → SharedExpense.
payerId (FK)	Integer	Required; → User; must be a participant of the expense.

A.7 Budget

Attribute	Type	Constraints / Description
budgetId	Integer	PK, auto-generated.
totalAmount	Decimal(12,2)	Required; 0.01–1,000,000.00; ≥ sum of its category limits.
period	String(7)	Required; format YYYY-MM; (userId, period) is unique.
userId (FK)	Integer	Required; → User.

A.8 CategoryBudgetLimit

Attribute	Type	Constraints / Description
limitId	Integer	PK, auto-generated.
limitAmount	Decimal(12,2)	Required; > 0; sum of limits per budget ≤ Budget.totalAmount.
budgetId (FK)	Integer	Required; → Budget.
categoryId (FK)	Integer	Required; → ExpenseCategory; (budgetId, categoryId) is unique.

A.9 Settlement

Attribute	Type	Constraints / Description
settlementId	Integer	PK, auto-generated.
amount	Decimal(12,2)	Required; > 0; ≤ payer's debt and ≤ receiver's credit at recording time.
settlementDate	DateTime	Set by the server when recorded (A-09).
status	Enum	Pending (default), Confirmed, Rejected. Only Confirmed affects balances.
sharedExpenseId (FK)	Integer	Required; → SharedExpense.
payerId (FK)	Integer	Required; → User (the debtor who settles).
receiverId (FK)	Integer	Required; → User; different from payerId; both must be participants.

A.10 BudgetNotification

Attribute	Type	Constraints / Description
notificationId	Integer	PK, auto-generated.
message	String(255)	Required; generated text, e.g., "You have used 82.5% of your Food limit for October 2026".
createdAt	DateTime	Set by the server.
thresholdReached	Integer	Required; 80 or 100.
isRead	Boolean	Default false.
budgetId (FK)	Integer	Required; → Budget.
limitId (FK)	Integer	Optional; → CategoryBudgetLimit (null for warnings on the monthly total).

A.11 Derived Values (calculated, not stored)

Value	Definition
Net balance	Amount paid − share + confirmed settlements sent − confirmed settlements received, per participant and shared expense (FR-12).
Shared expense status	Open or Settled (FR-12 rule 5).
Spent (month / category)	Sum of personal expenses in the month (and category) (FR-16).
Remaining	Budget or limit amount − spent.
Usage %	Spent ÷ budget or limit × 100, one decimal place.
Change %	(B − A) ÷ A × 100 between two months; “New” when A = 0 (FR-19).

Appendix B: Glossary

Terms already defined in Section 1.3 are not repeated here.

Term	Definition
95th percentile	The response time within which 95% of measured requests complete (NFR-01).
Creditor / Debtor	A participant with a positive net balance (is owed money) / a negative net balance (owes money).
Custom split	A split in which the user enters each participant’s share; the shares must equal the total.
Dashboard	The page shown after login, summarizing the month’s spending, budget usage, amounts owed, and unread notifications.
Equal split	A split that divides the total evenly among participants, with any remaining piasters added to the creator’s share.
HTTPS / TLS	HTTP secured with Transport Layer Security, which encrypts data between the browser and the server.
Net balance	See Appendix A.11.
ORM	Object-Relational Mapping: a library that maps Model classes to database tables and builds safe queries.
Password hash / salt	A one-way transformation of a password; the salt is random data added before hashing so identical passwords produce different hashes.
Pending settlement	A settlement recorded by a debtor and not yet confirmed or rejected by the receiver.
Piaster	One hundredth of an Egyptian pound (EGP 0.01), the smallest amount FinPact stores.
Pull request (PR)	A GitHub request to merge a member’s branch into the main branch after review.

Term	Definition
Session	The server-side record that keeps a user logged in between requests; it ends at logout or after 30 minutes of inactivity.
SQL injection	An attack that inserts SQL code through input fields; prevented by parameterized queries or an ORM.
Threshold	A usage level (80% or 100%) at which a budget warning is generated.
Viewport	The visible area of a web page in the browser window, in pixels.
Who-owes-whom list	The list, derived from net balances, showing which debtor should settle how much with which creditor.